

>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

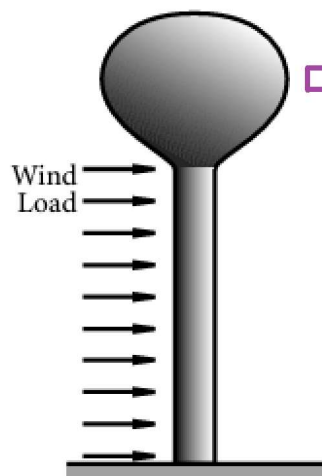
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A WATER TOWER: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)

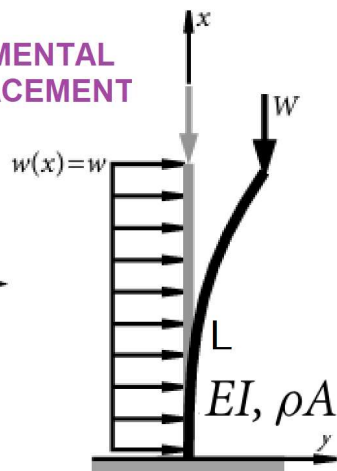
20
21
19
m
7

2
3
1
m
1

$$W(x, t) = P(t) = W_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$



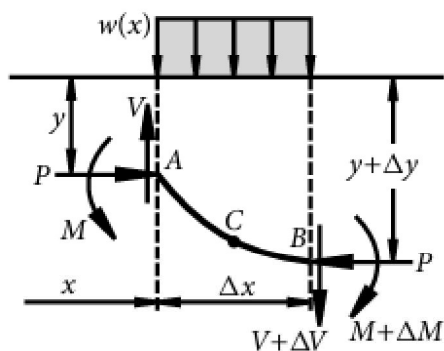
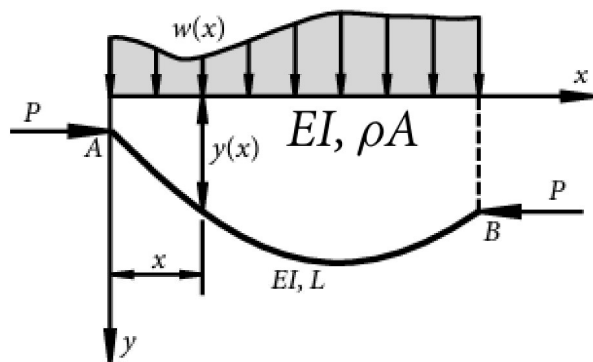
Water Tower



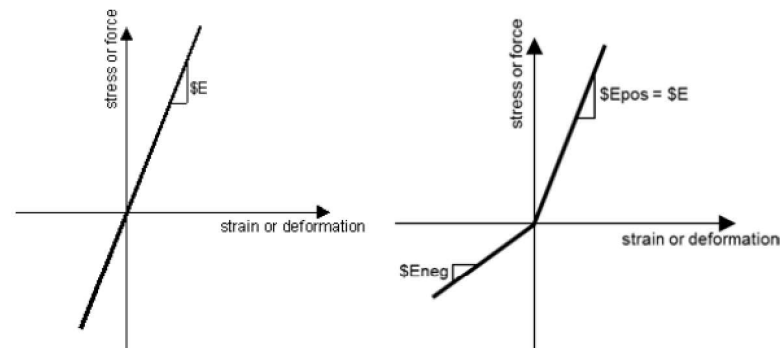
Beam-Column

Water tower.

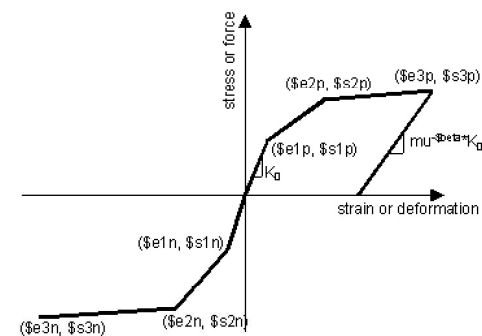
$$W(x, t) = P(t) = W_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$



Beam-column.



ELEMENT ELASTIC MATERIAL



ELEMENT INELASTIC MATERIAL

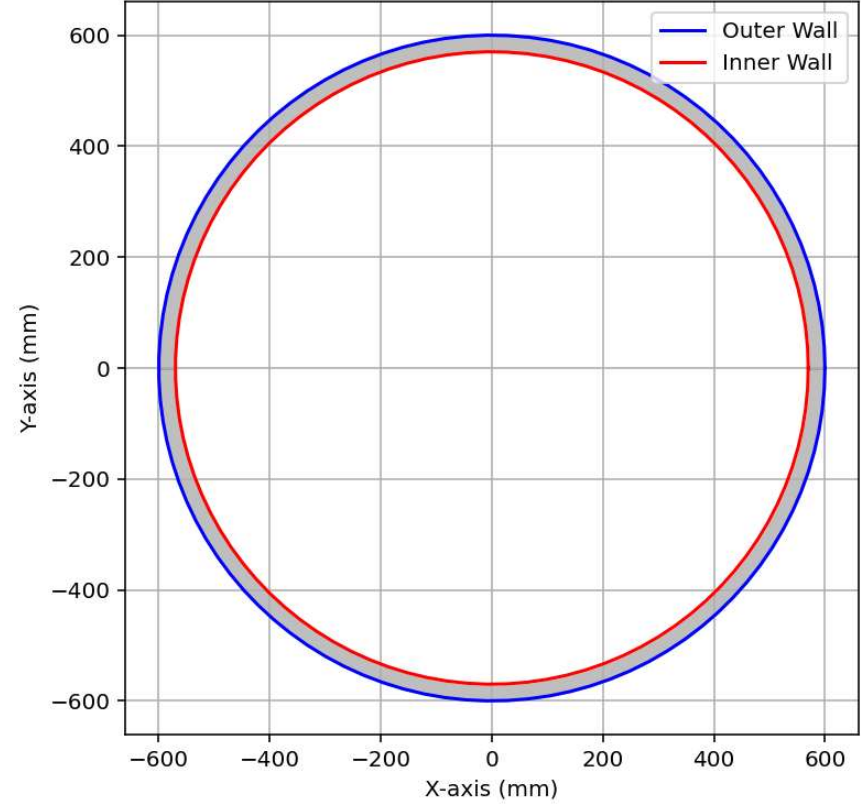
$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

Steel Pipe Hollow Section



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...X\++WATER_TOWER_8_ANA\W(x,t)_WATER_TOWER_8_ANA.py

W(x,t)_WATER_TOWER_8_ANA.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A WATER TOWER : AN OPENSEE
4 # FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-H
5 #
6 # W(x,t) = W0 sin(wt)
7 # W(x,t) = W0 exp(-0.05wt) sin(wt)
8 #
9 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
10 # OF ENERGY DISSIPATION CAPACITY INDEX
11 #
12 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
13 # EMAIL: salar.d.ghashghaei@gmail.com
14 #####
15 """
16 Nonlinear Seismic Performance Assessment of a Water Tower:
17 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
18
19 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Water
20 for performance-based earthquake engineering. The 2D model incorporates both material non
21 (elastic-perfectly plastic or hysteretic steel with strain hardening)
22 and geometric nonlinearity to capture P-delta effects
23 and large displacements—essential for collapse assessment.
24
25 Eight analysis protocols are implemented:
26 (1) [PERIOD] : Structural Period
27 (2) [STATIC] : Gravity load analysis establishing dead load state
28 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
29 and plastic mechanism identification
30 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
31 pinching behavior, and energy dissipation degradation
32 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
33 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
34 (7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
```

Steel Pipe Hollow Section

Y-axis (mm)

X-axis (mm)

Outer Wall

Inner Wall

Python Console Files Help Variable Explorer Debugger History Plots

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 817, Col 18 UTF-8 CRLF RW Mem 37%

STATIC ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...X\++WATER_TOWER_8_ANA\W(x,t)_WATER_TOWER_8_ANA.py

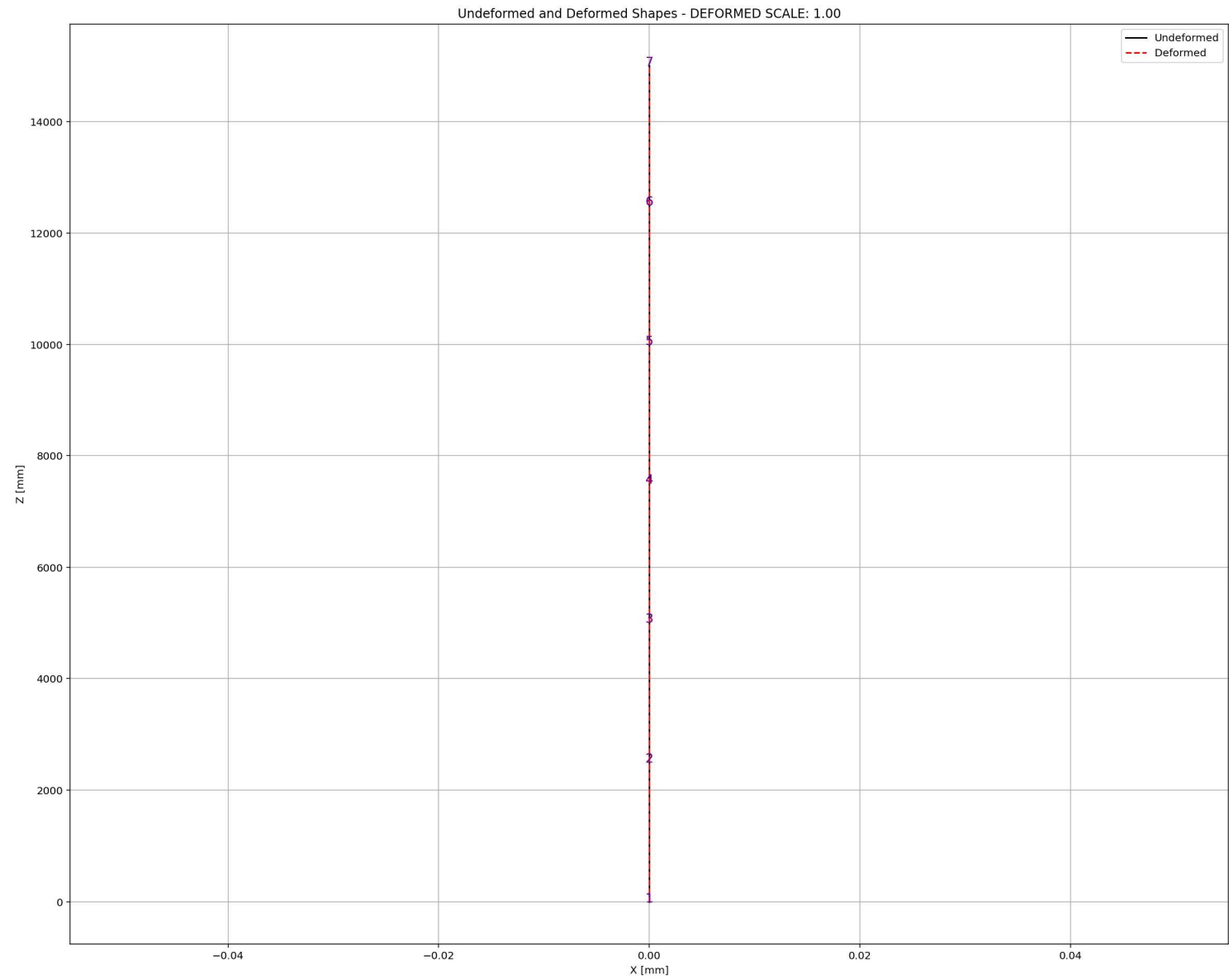
W(x,t)_WATER_TOWER_8_ANA.py X

```
1036 #####
1037 LENGTH = 15000.0 # [mm] Beam Length
1038 TOTAL MASS = 0.0 # [kg] Total Mass of Structure
1039 #####
1040 # PERIOD ANALYSIS
1041 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1042 #ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1043 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1044 ANAL_TYPE = 'PERIOD'
1045
1046 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1047 (PERIOD_MIN_X, PERIOD_MAX_X) = DATA
1048 print('Structure First Period: ', PERIOD_MIN_X)
1049 print('Structure Second Period: ', PERIOD_MAX_X)
1050
1051 S01.PLOT_2D FRAME(deformed_scale=1.0)
1052 #####
1053 # STATIC ANALYSIS
1054 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1055 #ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1056 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1057 ANAL_TYPE = 'STATIC'
1058
1059 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1060 (reactionX, reactionY, reactionZ, disp_mid,
1061 ele_axialforce, ele_shearforce, ele_momentforce,
1062 node_displacementsX, node_displacementsY,
1063 dispX, dispY) = DATA
1064
1065 S01.PLOT_2D FRAME(deformed_scale=1.0)
1066 #####
1067 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1068 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1069 #ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
```

Untransformed and Deformed Images - (DOF CHANNEL SCALE: 1.00)

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PUSHOVER ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...X\++WATER_TOWER_8_ANA\W(x,t)_WATER_TOWER_8_ANA.py

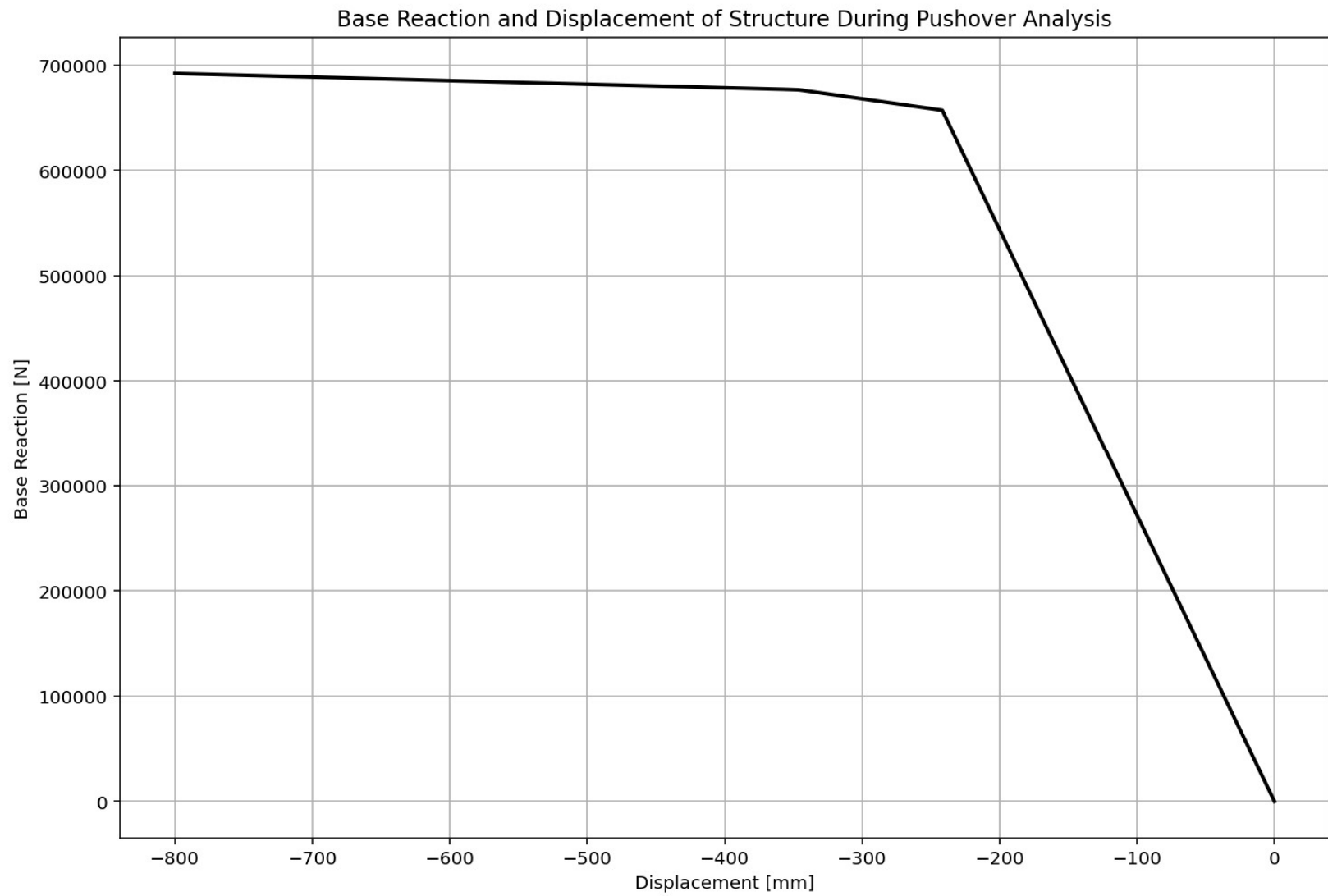
W(x,t)_WATER_TOWER_8_ANA.py X

```
1067 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1068 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1069 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1070 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1071 ANAL_TYPE = 'PUSHOVER'
1072
1073 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1074 (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
1075 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1076 node_displacementsX_PUSH, node_displacementsY_PUSH,
1077 dispX_PUSH, dispY_PUSH,
1078 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1079 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINT_PUSH, CURVATURE_PUSH) = DATA
1080
1081
1082 XDATA = disp_mid_PUSH
1083 YDATA = reactionX_PUSH
1084 XLABEL = 'Displacement [mm]'
1085 YLABEL = 'Base Reaction [N]'
1086 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1087 COLOR = 'black'
1088 SEMILOGY = False
1089 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1090
1091 DATA = S07.BILINEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionX_PUSH), SLOPE_NODE=10)
1092 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor)
1093
1094 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1095 plt.figure(0, figsize=(12, 8))
1096 plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1097 plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1098 plt.title('Period of Structure During Pushover Analysis')
1099 plt.ylabel('Structural Period [s]')
1100 plt.xlabel('Displacement [mm]')
```

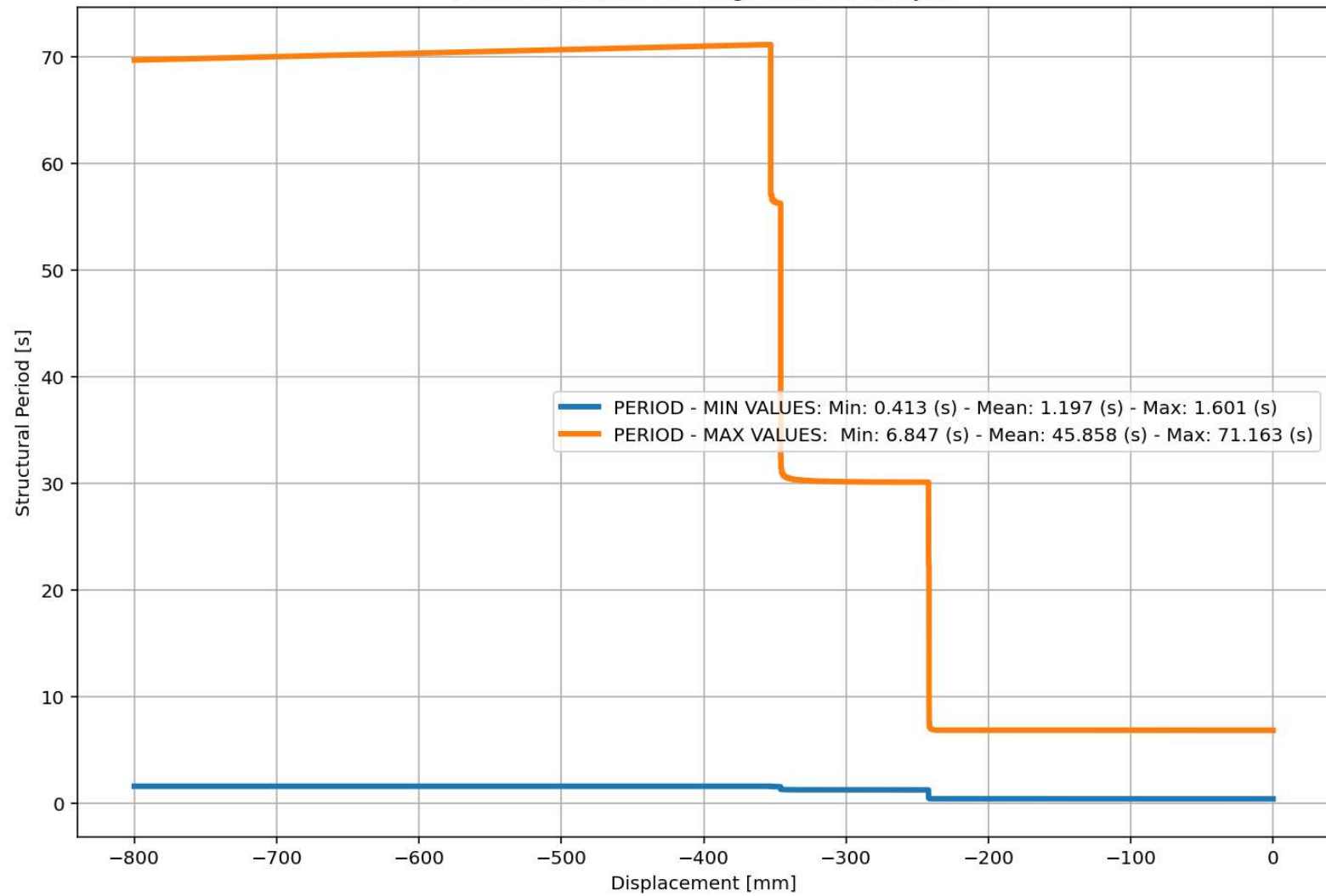
Base Reaction and Displacement of Structure During Pushover Analysis

IPython Console Files Help Variable Explorer Debugger History Plots

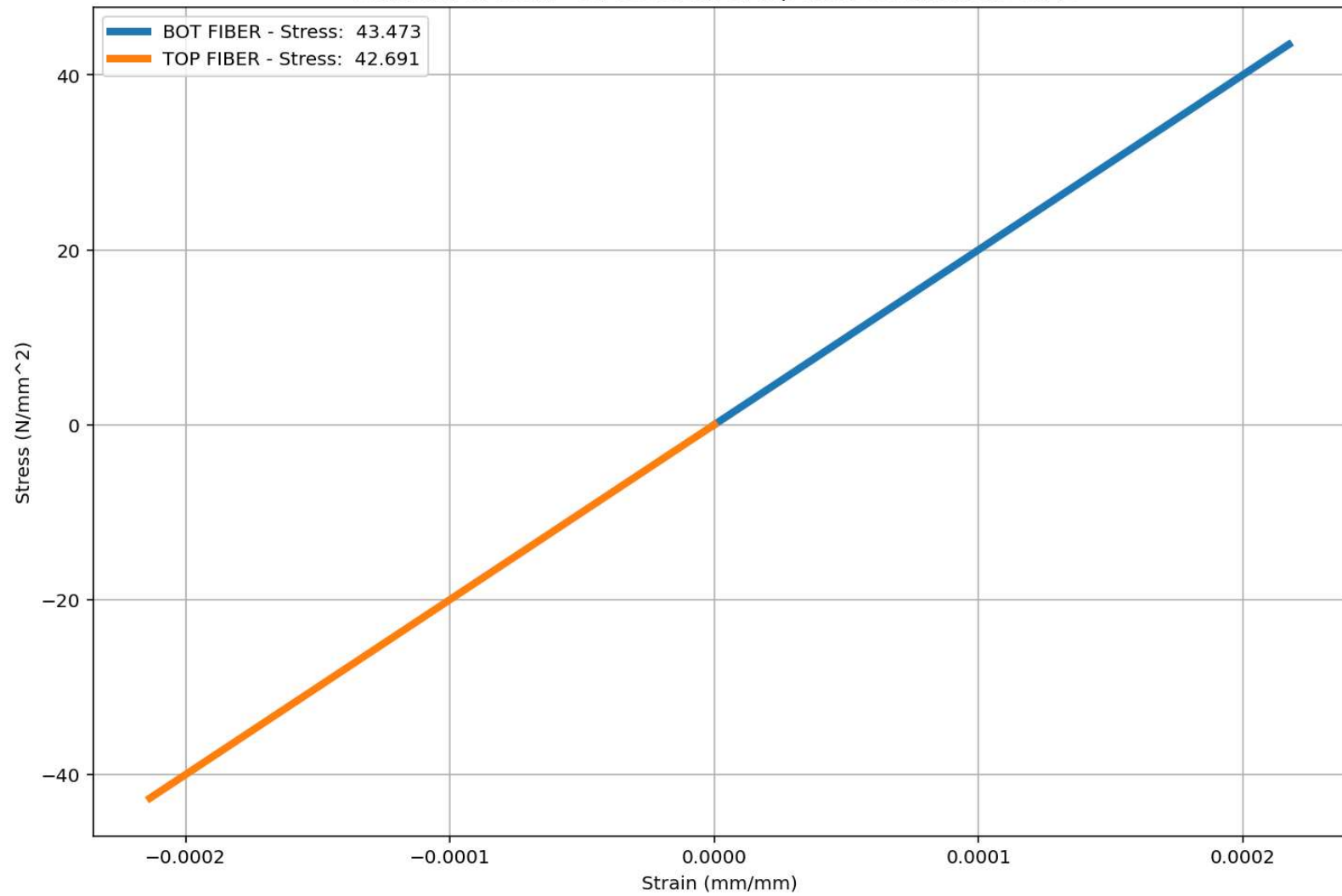
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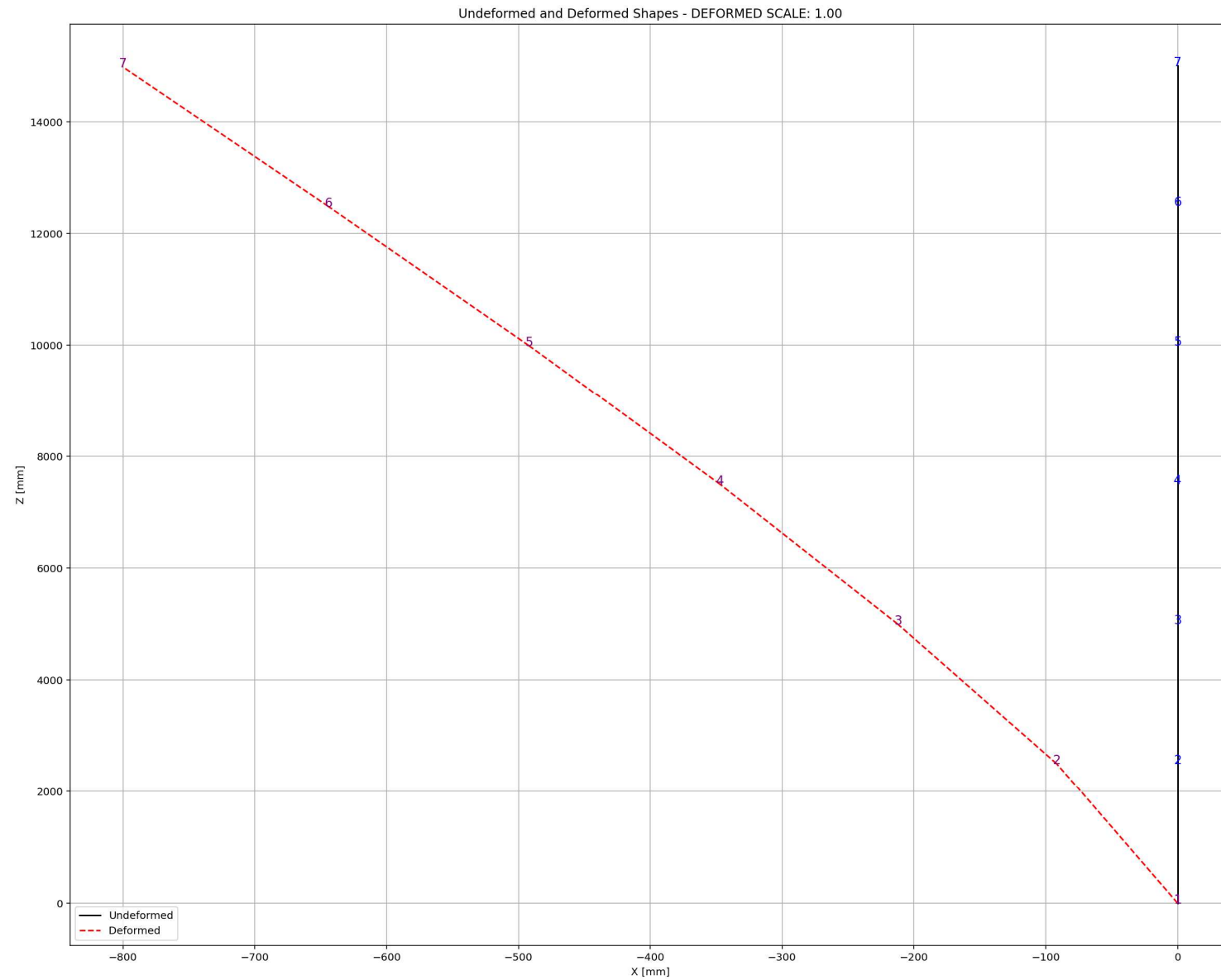


Period of Structure During Pushover Analysis



Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

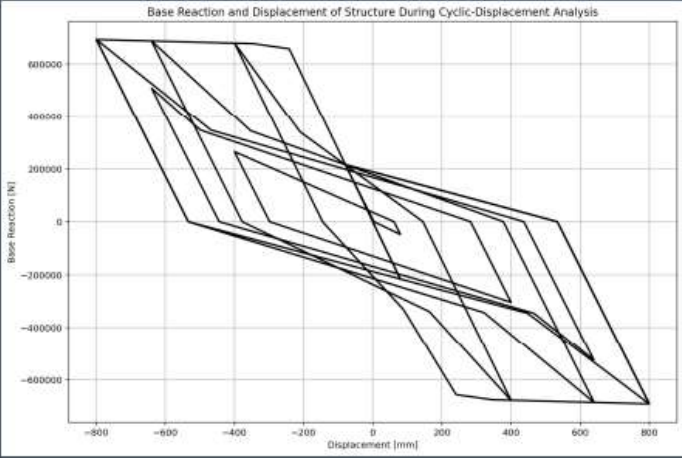
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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...X\++WATER_TOWER_8_ANA\W(x,t)_WATER_TOWER_8_ANA.py

W(x,t)_WATER_TOWER_8_ANA.py X

```
1121 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1122 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1123 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1124 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1125 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1126
1127 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1128 (reactionX_CP, reactionY_CP, reactionZ_CP, disp_mid_CP,
1129 ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1130 node_displacementsX_CP, node_displacementsY_CP,
1131 dispX_CP, dispY_CP,
1132 PERIOD_MIN_CP, PERIOD_MAX_CP,
1133 STRESSb_CP, STRAINb_CP, STRESSt_CP, STRAINt_CP, CURVATURE_CP) = DATA
1134
1135
1136 XDATA = disp_mid_CP
1137 YDATA = reactionX_CP
1138 XLABEL = 'Displacement [mm]'
1139 YLABEL = 'Base Reaction [N]'
1140 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1141 COLOR = 'black'
1142 SEMILOGY = False
1143 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1144
1145 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1146 plt.figure(0, figsize=(12, 8))
1147 plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
1148 plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
1149 plt.title('Period of Structure During Cyclic Displacement Analysis')
1150 plt.ylabel('Structural Period [s]')
1151 plt.xlabel('Displacement [mm]')
1152 #plt.semilogy()
1153 plt.grid()
1154 plt.legend(['f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(P
```

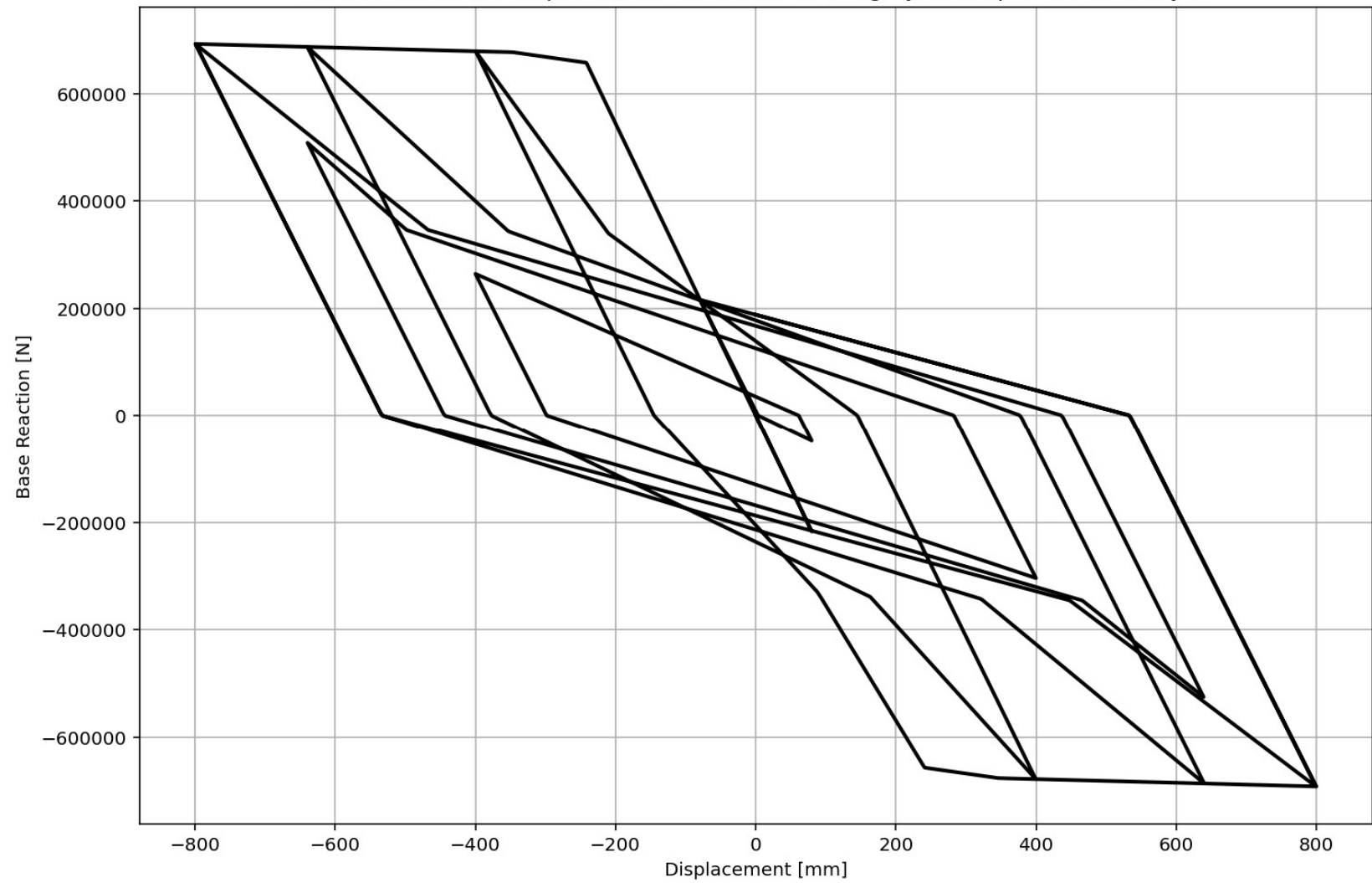
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis



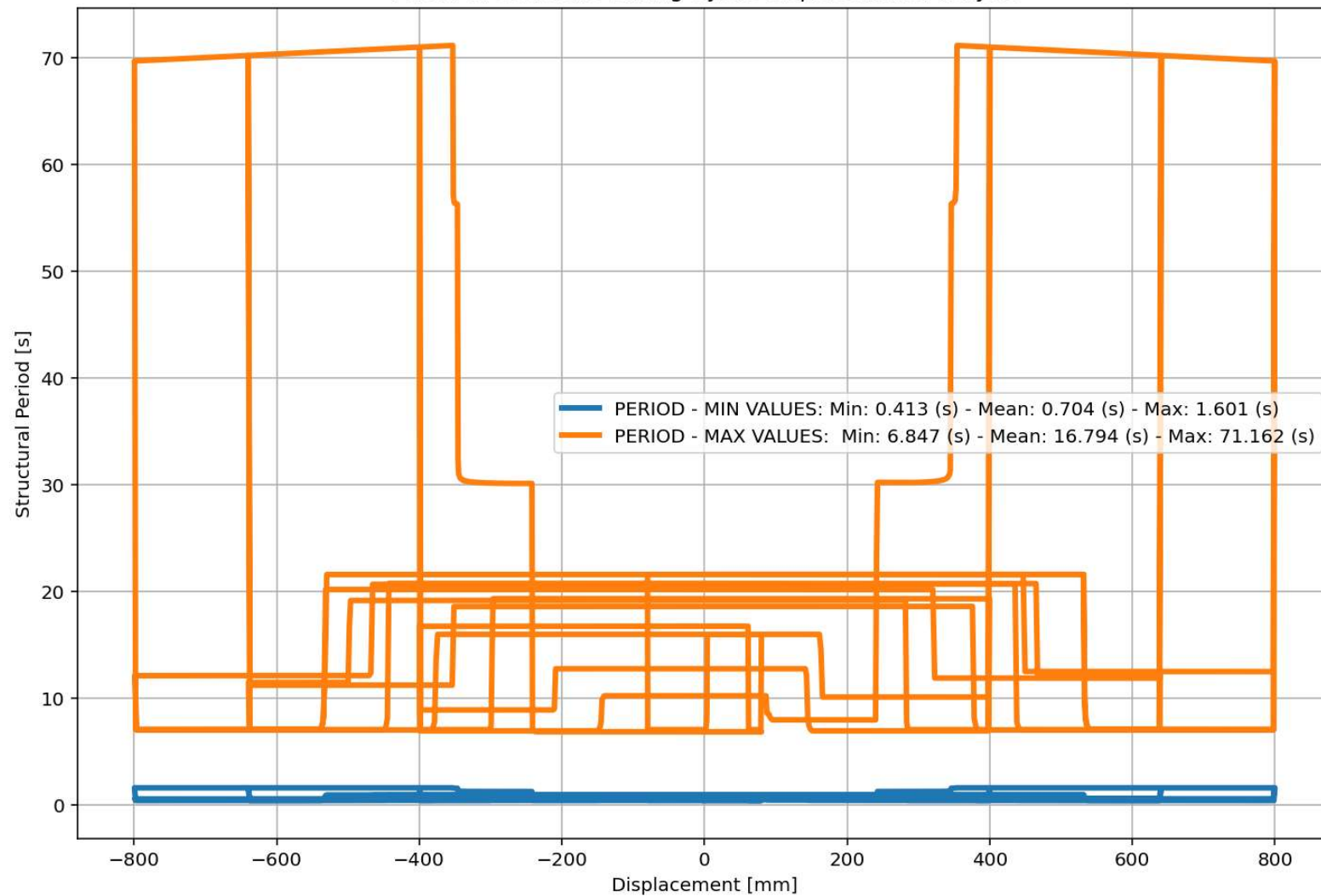
IPython Console Files Help Variable Explorer Debugger History Plots

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 817, Col 18 UTF-8 CRLF RW Mem 45%

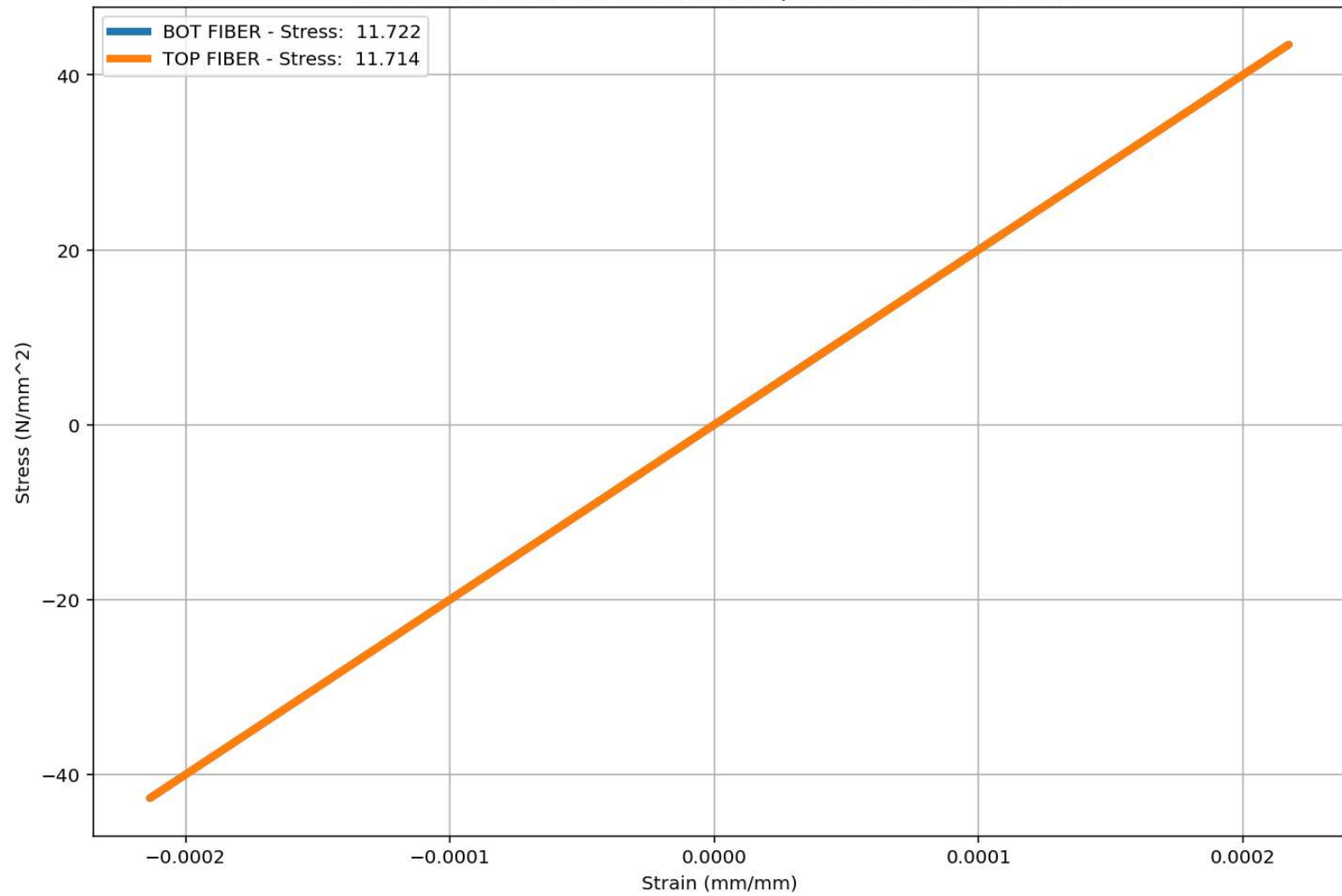
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis



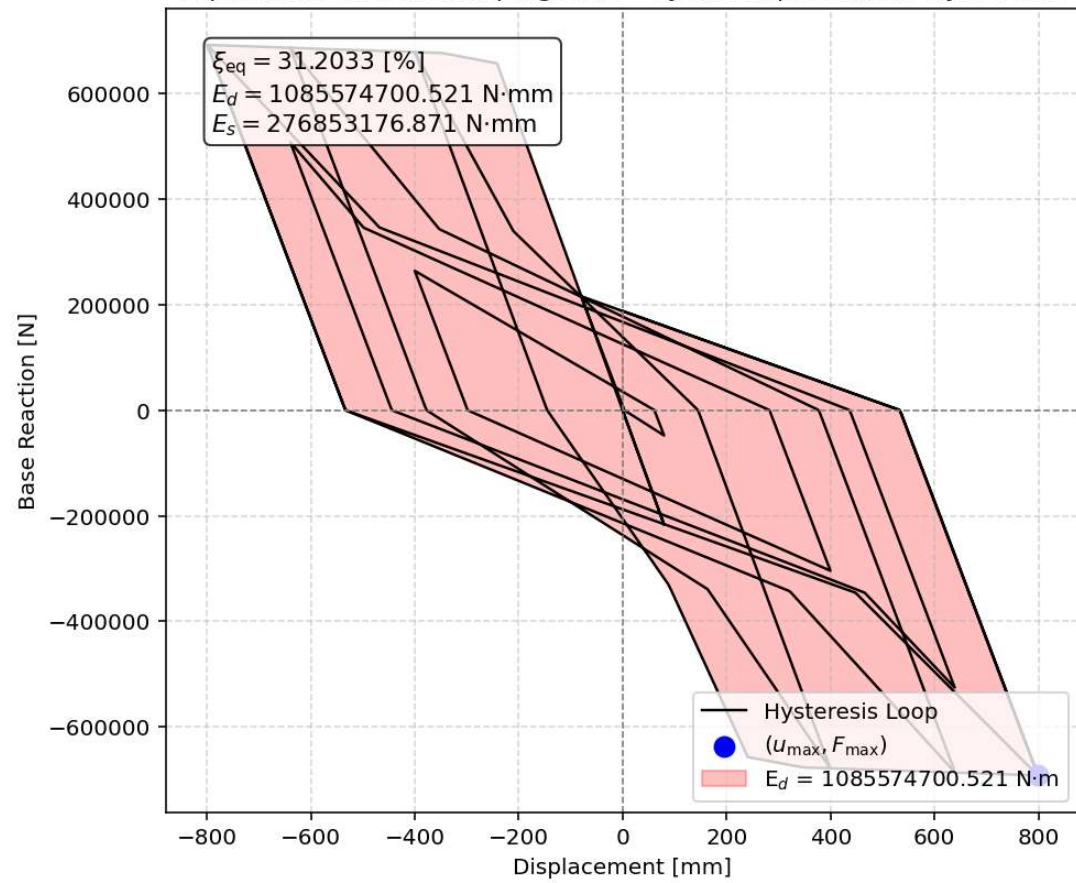
Period of Structure During Cyclic Displacement Analysis

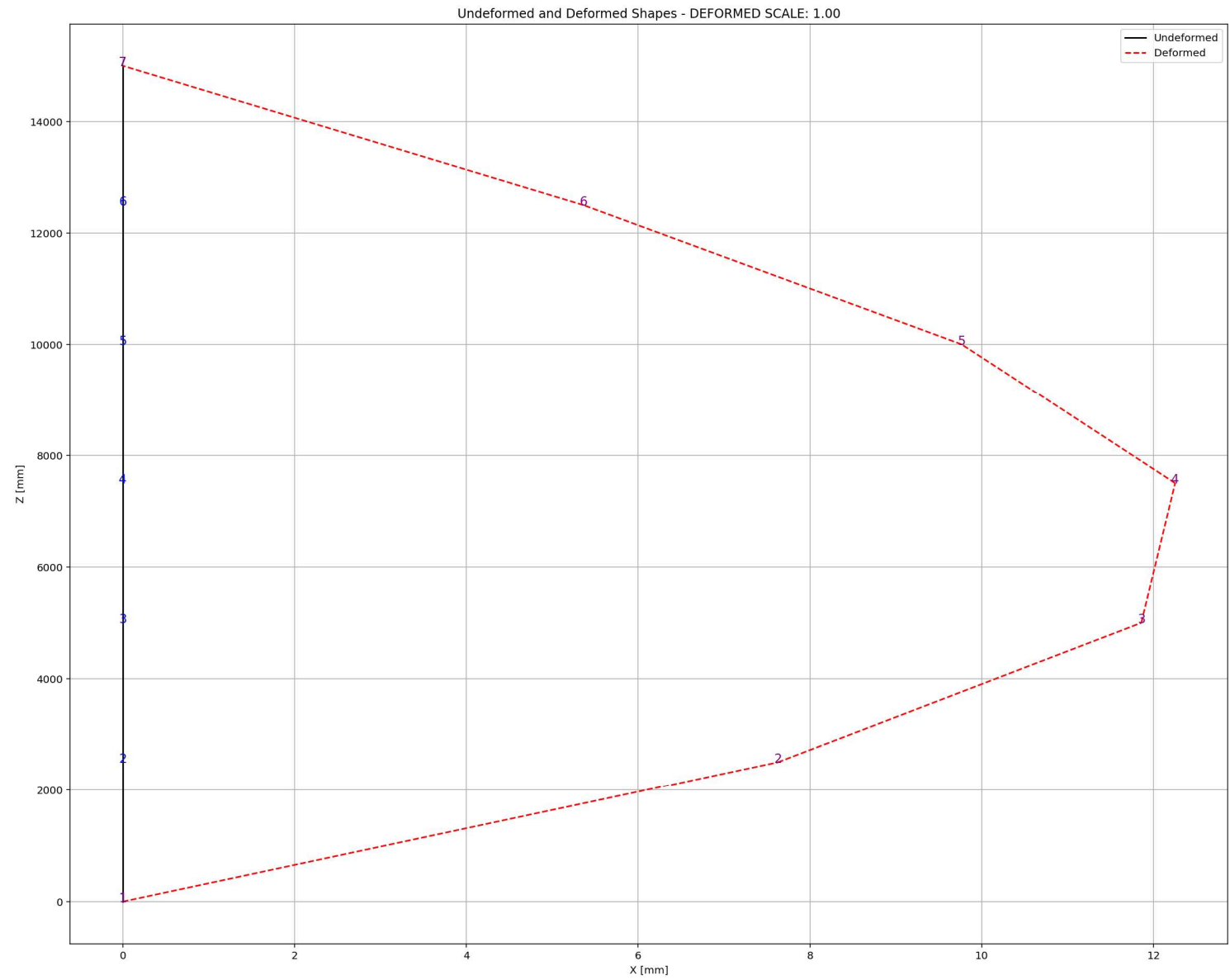


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis





STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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W(x,t)_WATER_TOWER_8_ANA.py X

```
1172 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1173 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1174 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1175 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1176 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1177
1178 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1179
1180 (reactionX_ETDLS, reactionY_ETDLS, reactionZ_ETDLS, disp_mid_ETDLS,
1181 ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1182 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1183 dispX_ETDLS, dispY_ETDLS,
1184 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1185 STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDLS) = DATA
1186
1187
1188 XDATA = disp_mid_ETDLS
1189 YDATA = reactionX_ETDLS
1190 XLABEL = 'Displacement [mm]'
1191 YLABEL = 'Base Reaction [N]'
1192 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
1193 COLOR = 'black'
1194 SEMIOLOGY = False
1195 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1196
1197 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1198 plt.figure(0, figsize=(12, 8))
1199 plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1200 plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1201 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1202 plt.ylabel('Structural Period [s]')
1203 plt.xlabel('Displacement [mm]')
1204 #plt.semilogy()
1205 plt.grid()
```

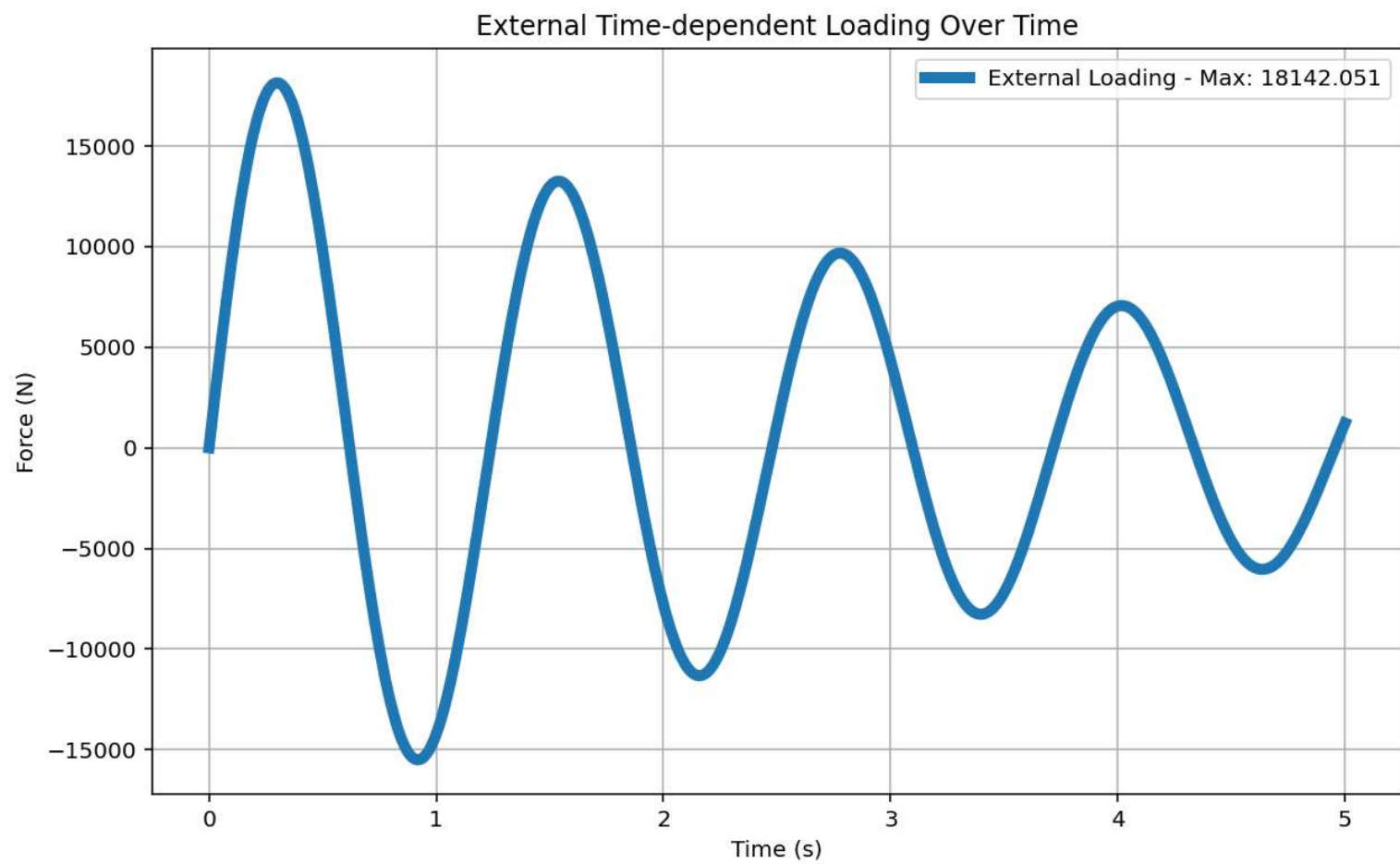
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

Base Reaction [N]

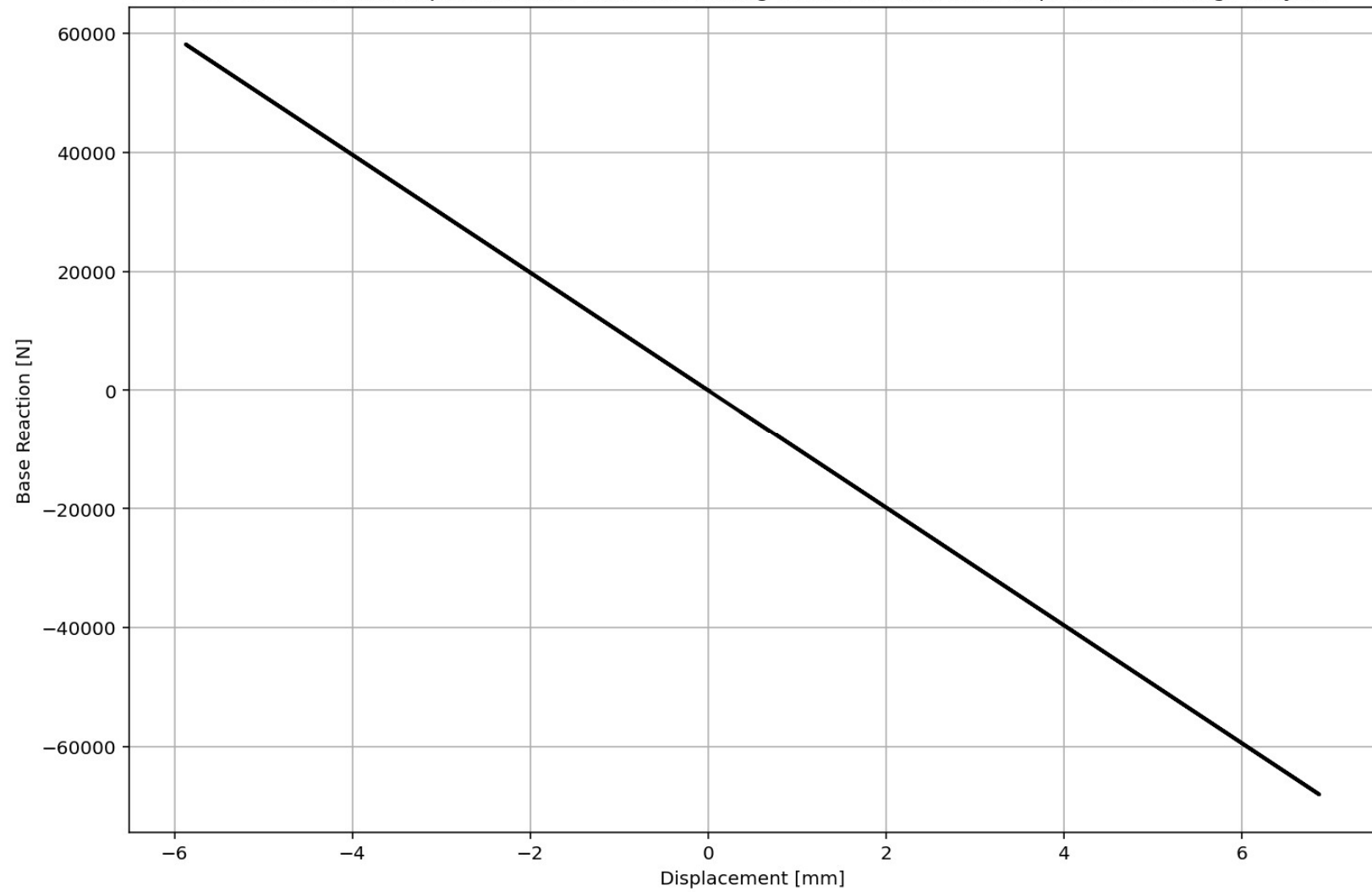
Displacement [mm]

Python Console Files Help Variable Explorer Debugger History Plots

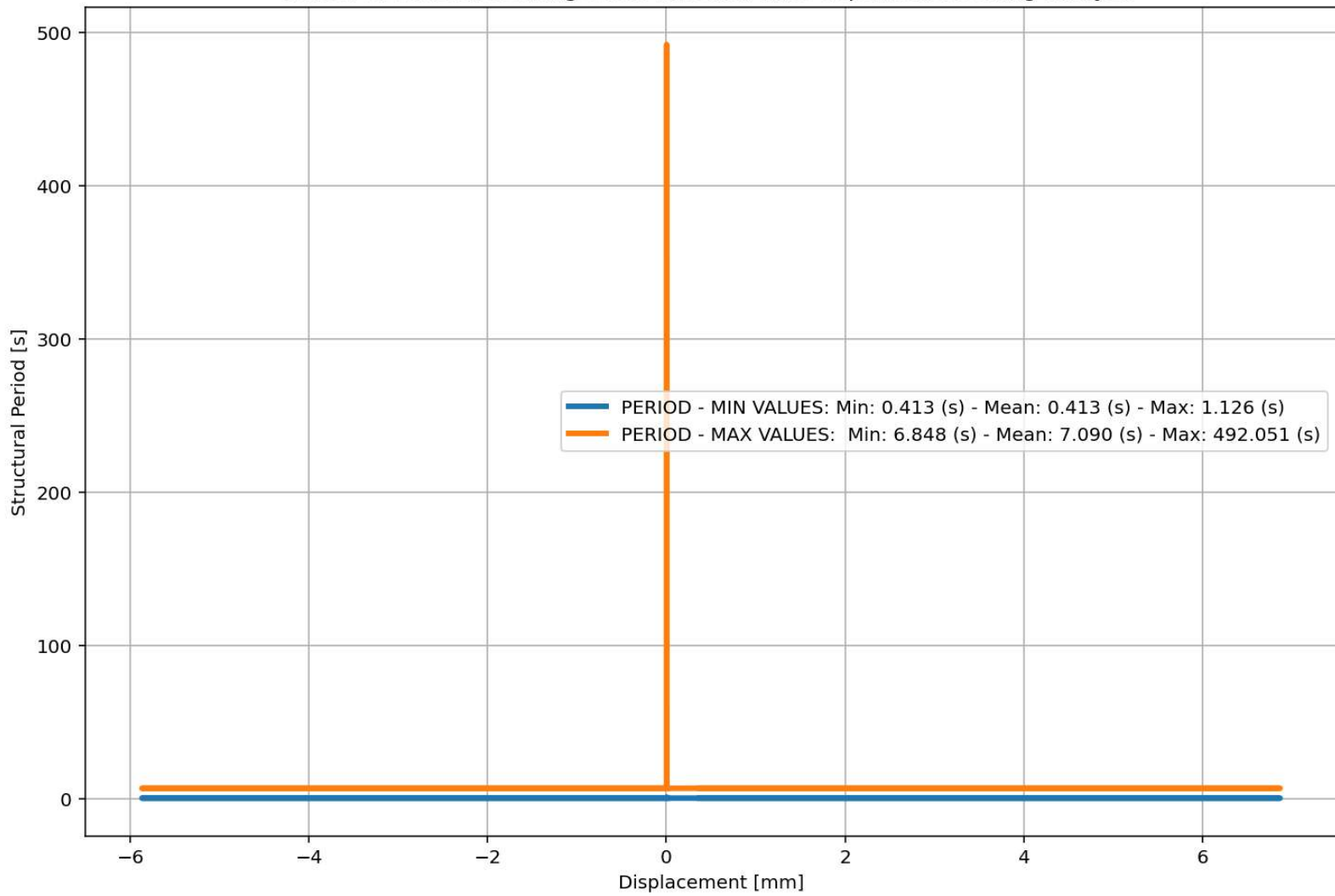
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 817, Col 18 UTF-8 CRLF RW Mem 45%



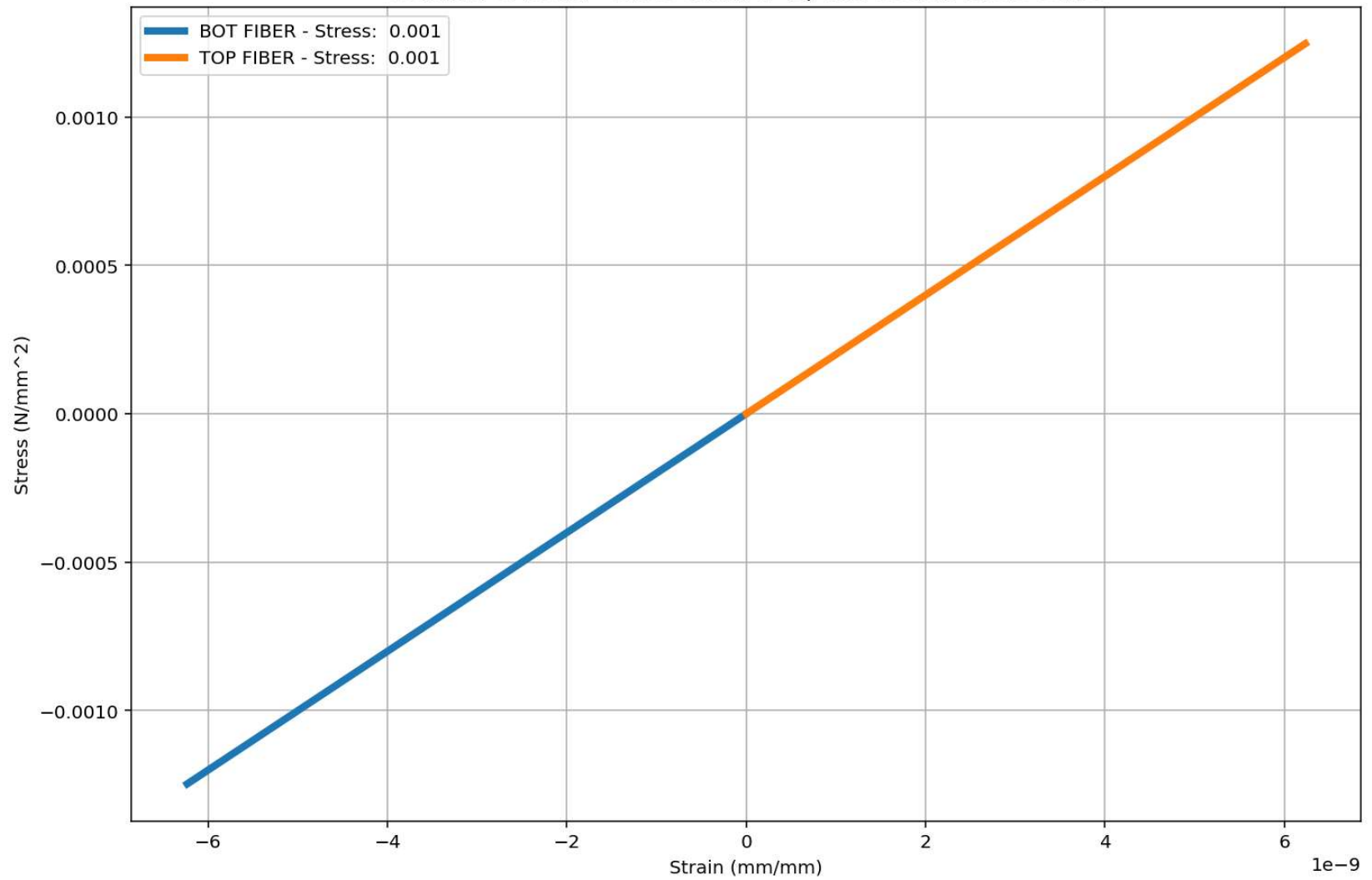
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

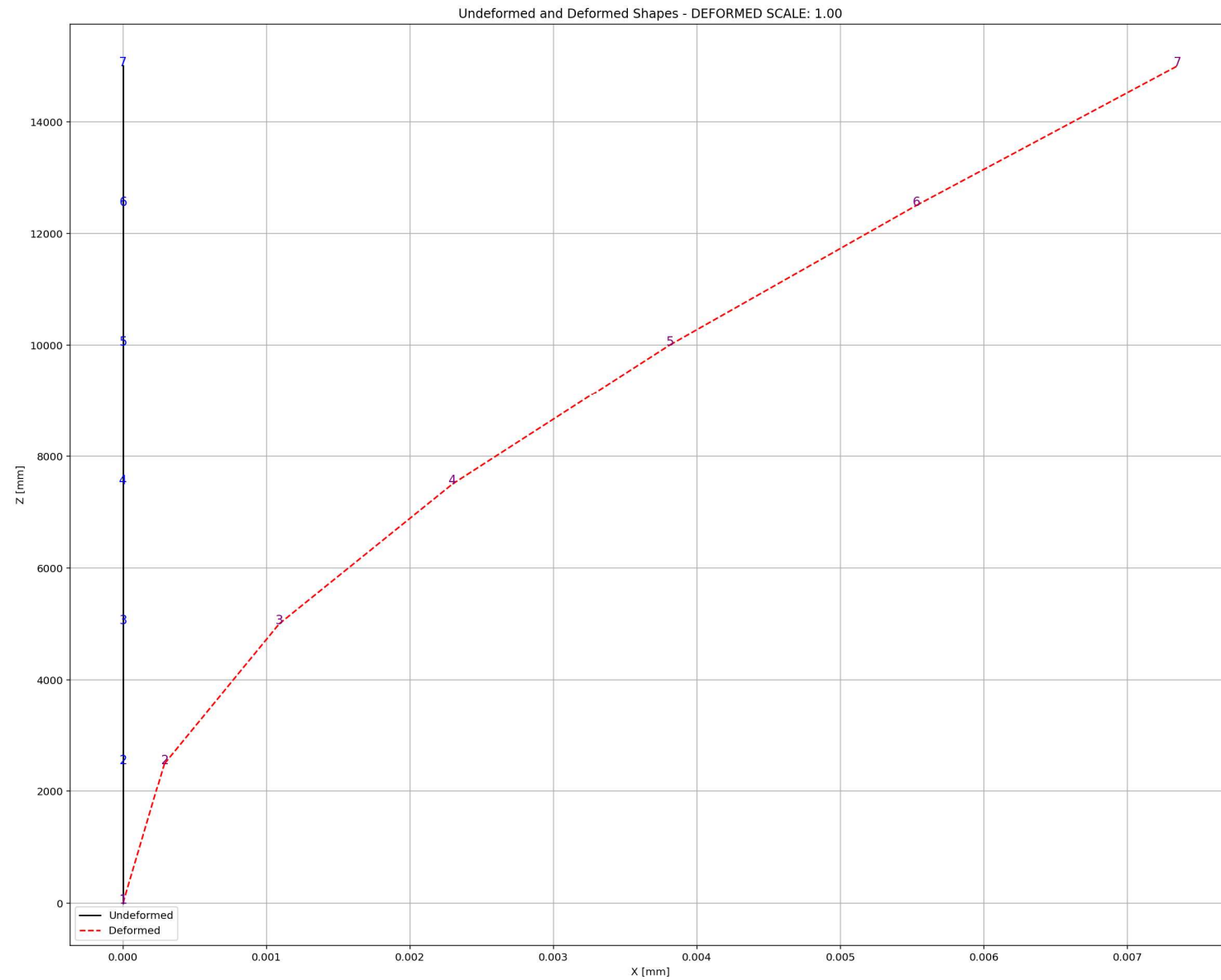


Period of Structure During Static External Time-dependent Loading Analysis



Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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W(x,t)_WATER_TOWER_8_ANA.py X

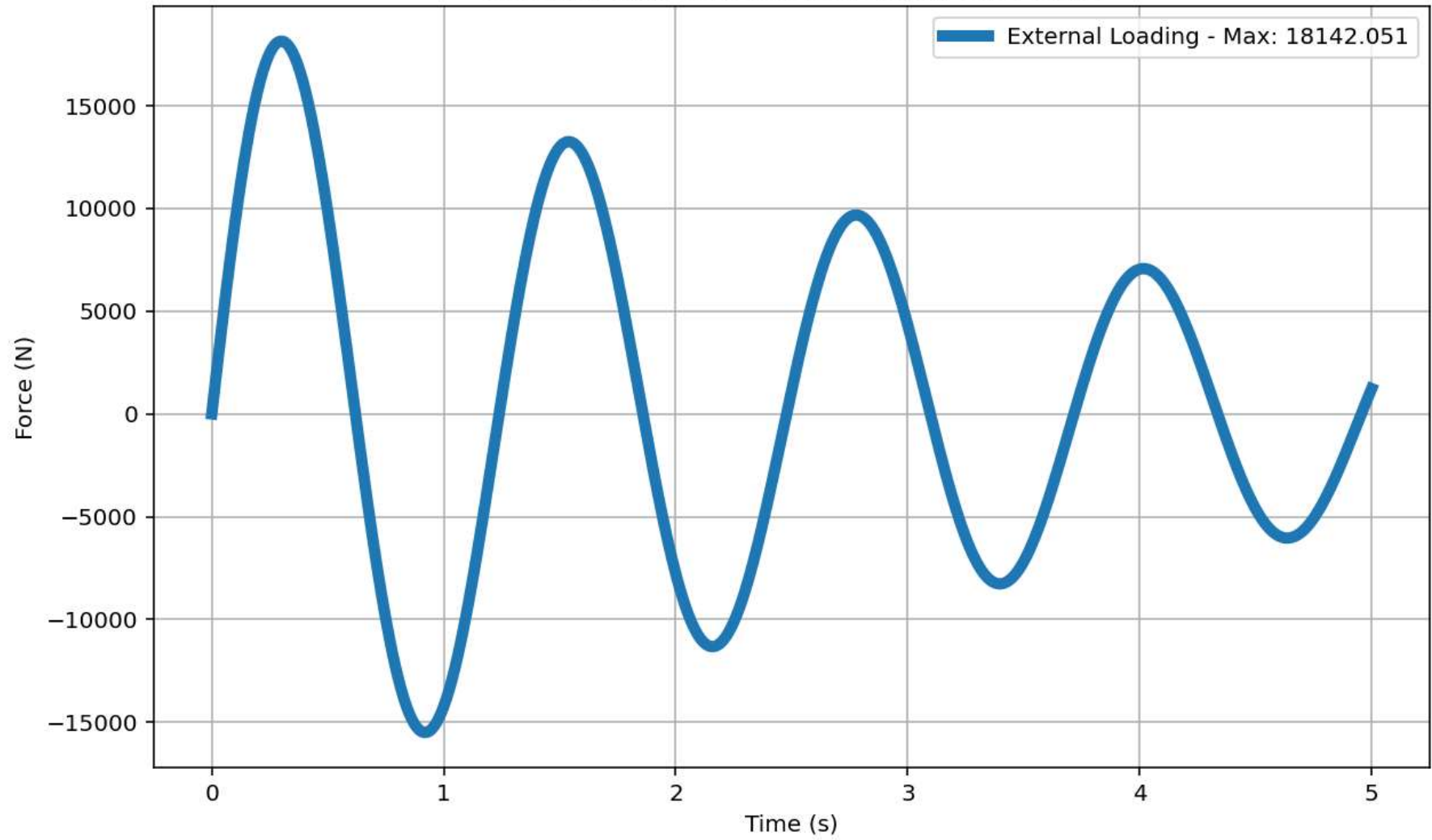
```
1224 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1225 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1226 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1227 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1228 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1229
1230 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1231
1232 (time_ETDLD, reactionX_ETDLD, reactionY_ETDLD, reactionZ_ETDLD, disp_mid_ETDLD,
1233 ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
1234 node_displacementsX_ETDLD, node_displacementsY_ETDLD,
1235 dispX_ETDLD, dispY_ETDLD,
1236 veloX_ETDLD, veloY_ETDLD,
1237 accX_ETDLD, accY_ETDLD,
1238 stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
1239 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
1240 STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINt_ETDLD, CURVATURE_ETDLD) = DATA
1241
1242
1243 XDATA = disp_mid_ETDLD
1244 YDATA = reactionX_ETDLD
1245 XLABEL = 'Displacement [mm]'
1246 YLABEL = 'Base Reaction [N]'
1247 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
1248 COLOR = 'black'
1249 SEMILOGY = False
1250 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1251
1252 PLOT_TIME_HISTORY(time_ETDLD, reactionX_ETDLD, disp_mid_ETDLD,
1253                   dispX_ETDLD, dispY_ETDLD,
1254                   veloX_ETDLD, veloY_ETDLD,
1255                   accX_ETDLD, accY_ETDLD)
1256
1257 # PLOT ELEMENTS AXIAL FORCE
```

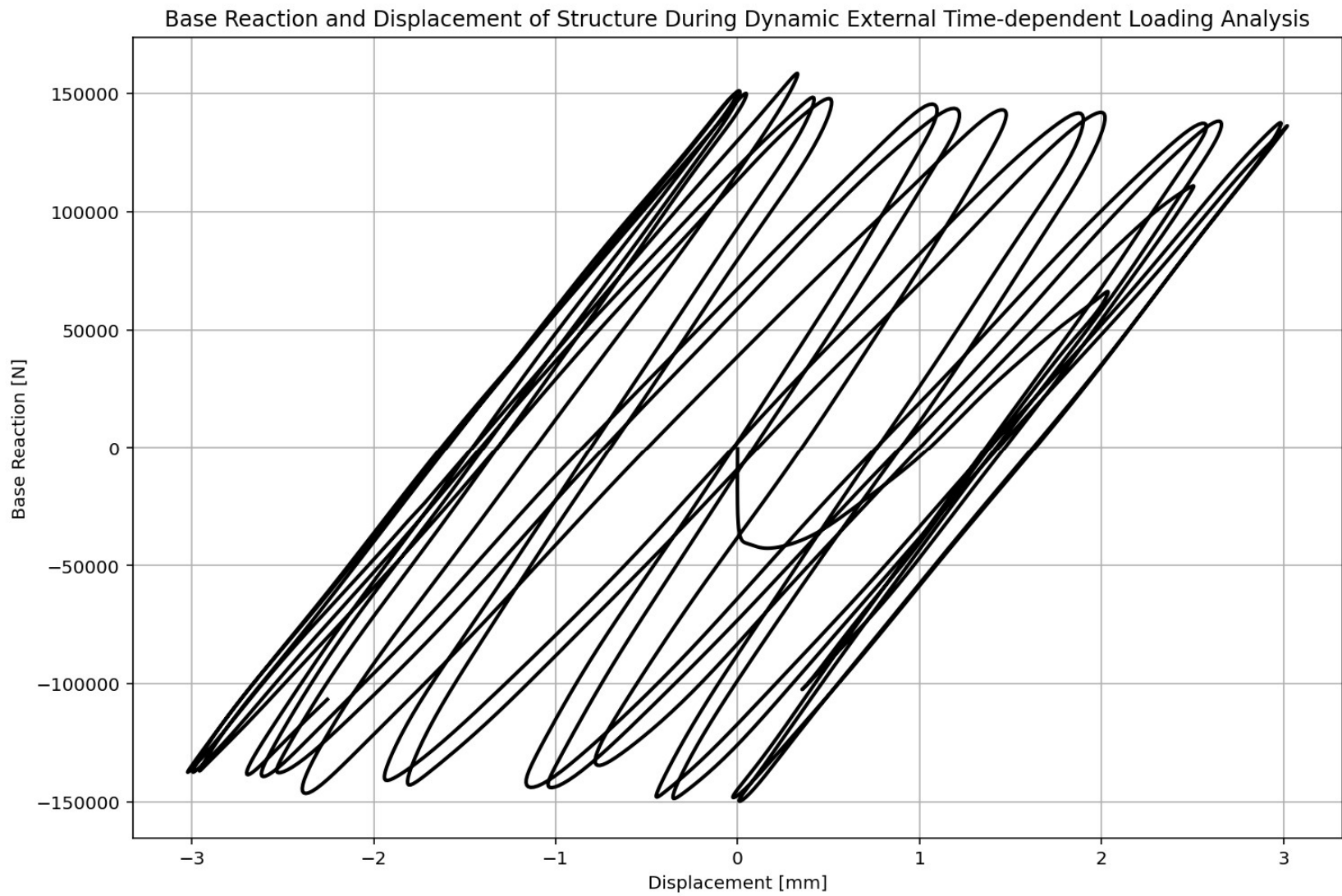
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis

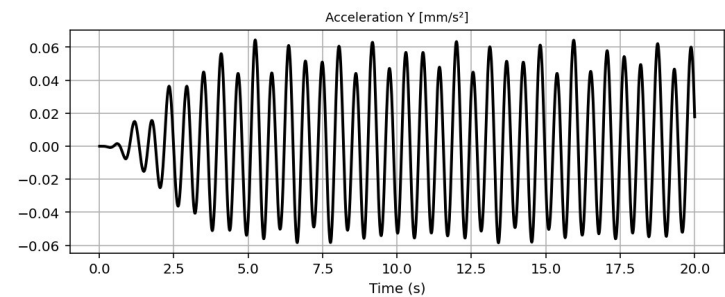
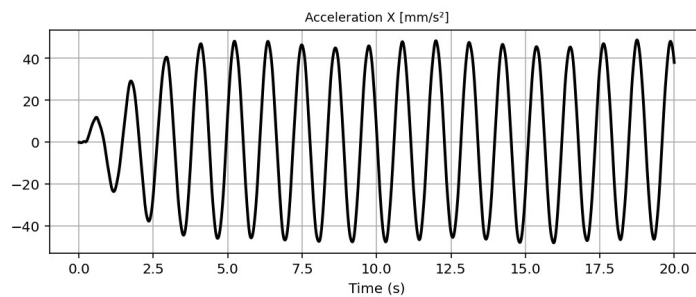
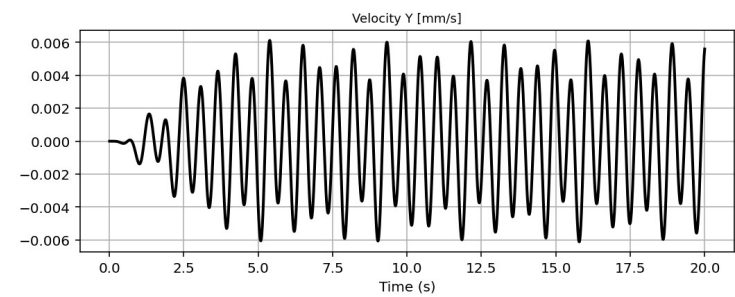
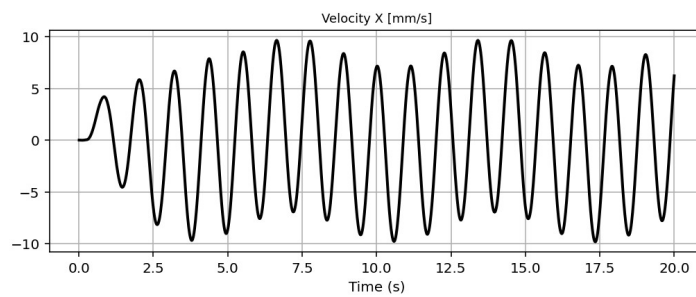
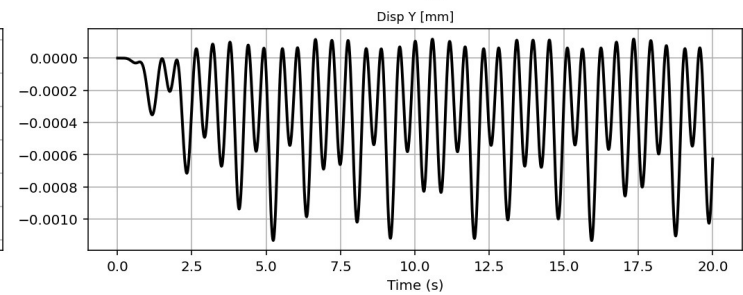
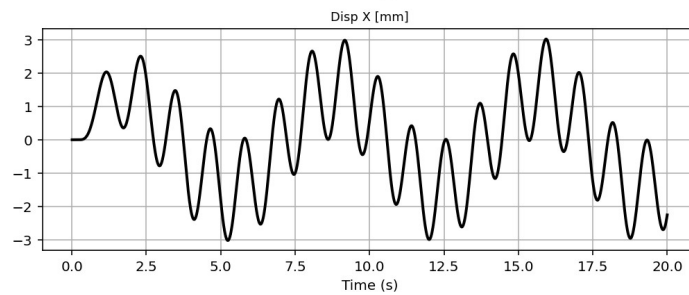
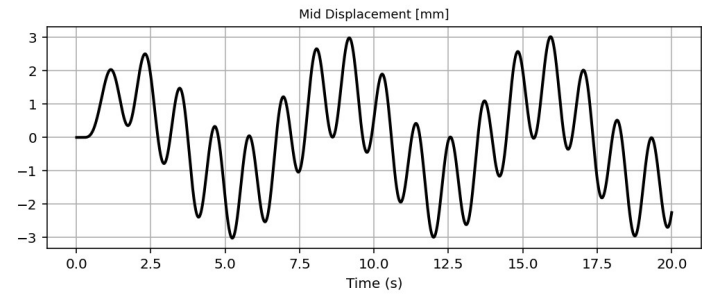
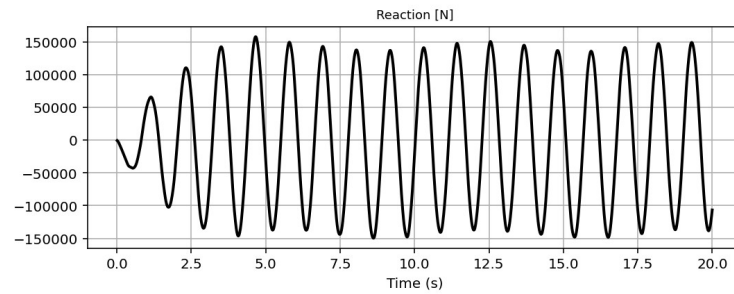
IPython Console Files Help Variable Explorer Debugger History Plots

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 817, Col 18 UTF-8 CRLF RW Mem 46%

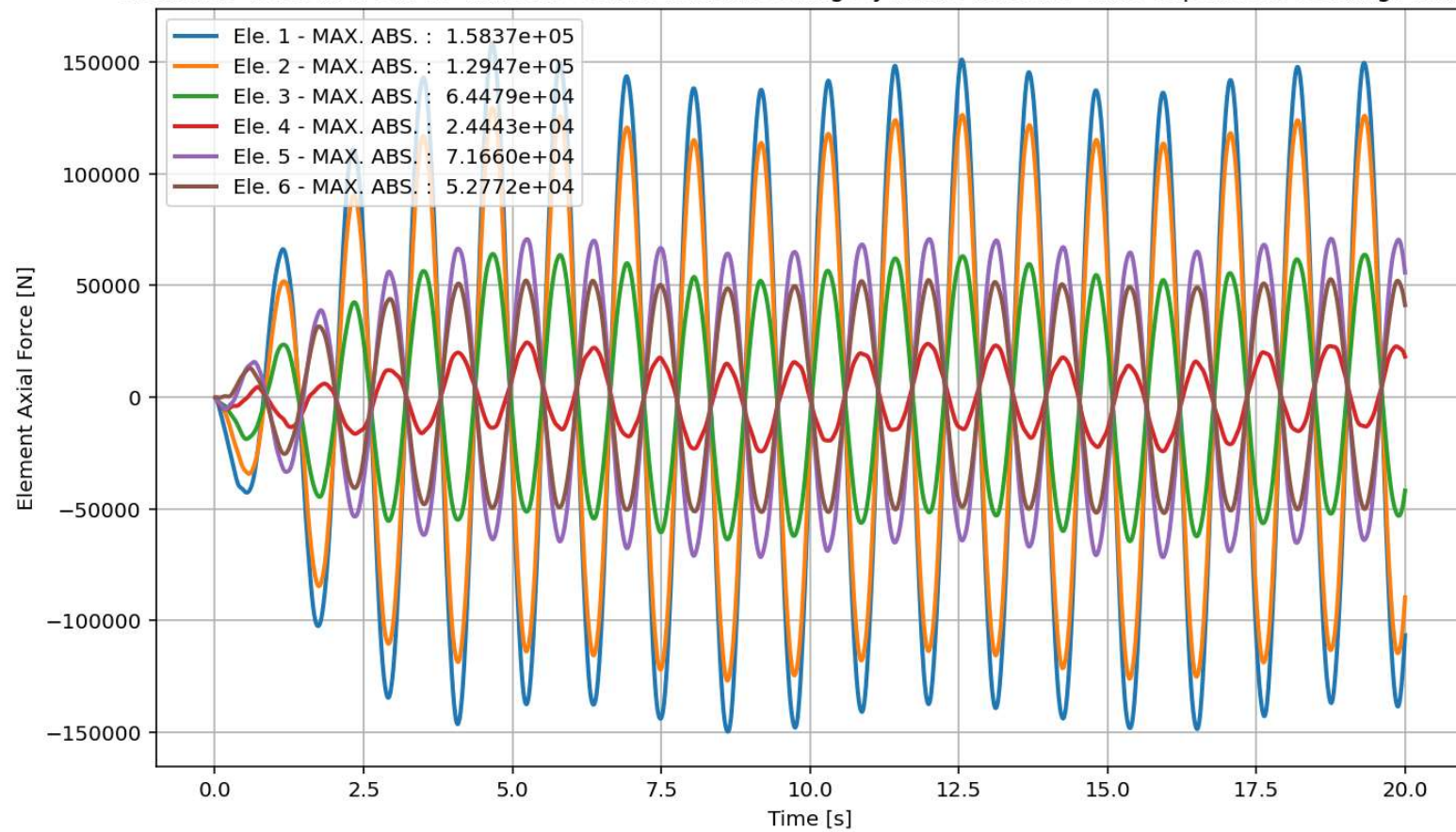
External Time-dependent Loading Over Time



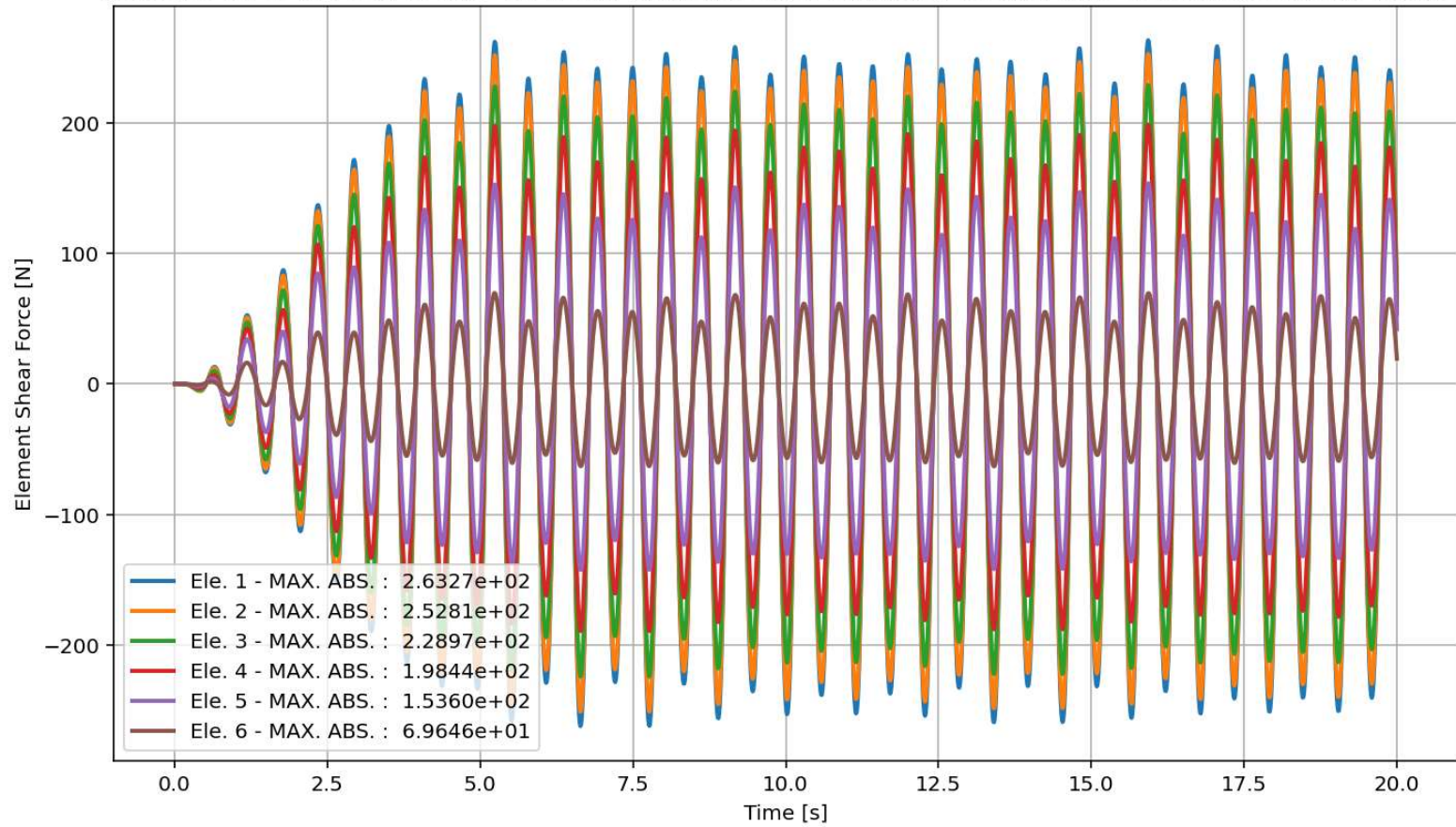




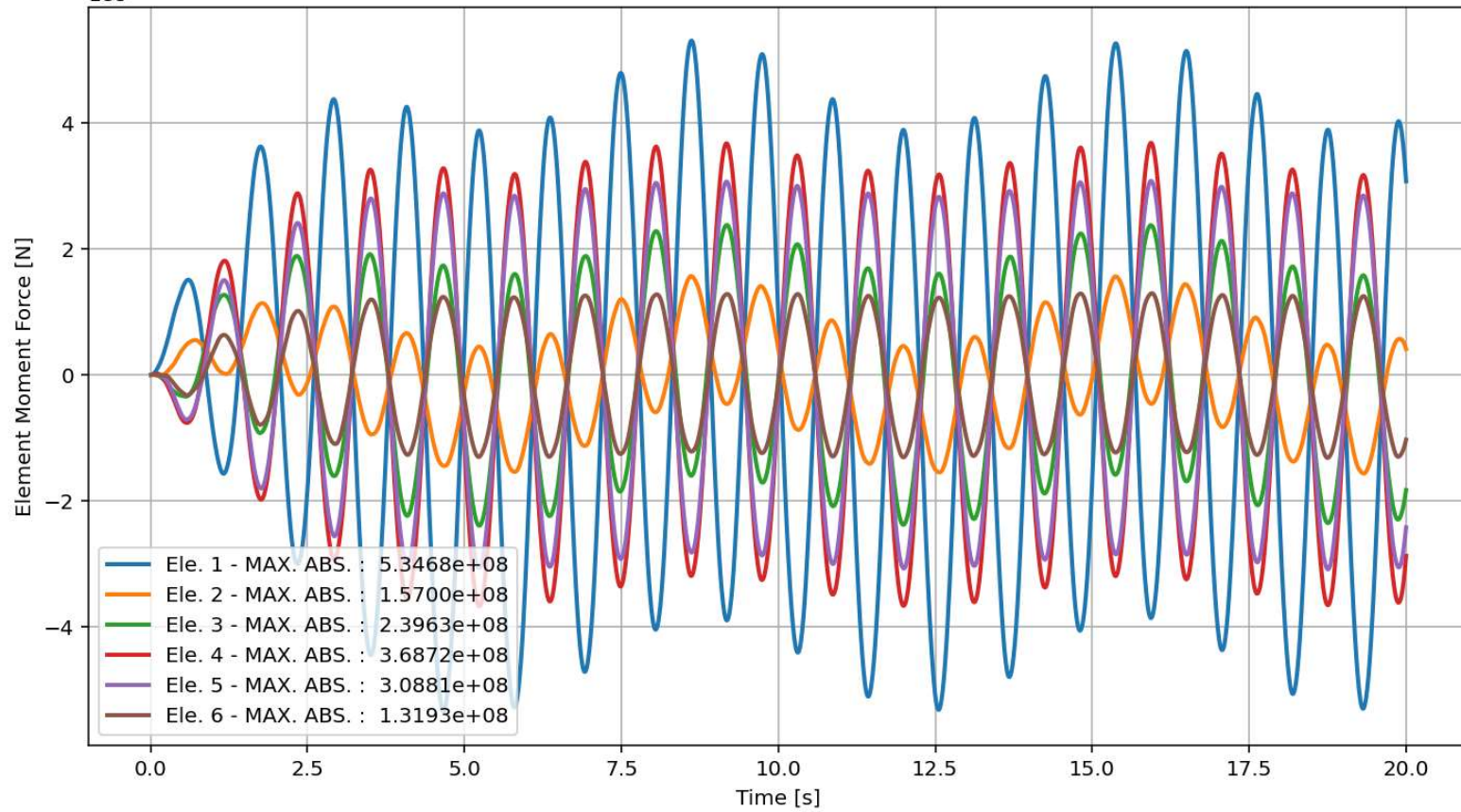
elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



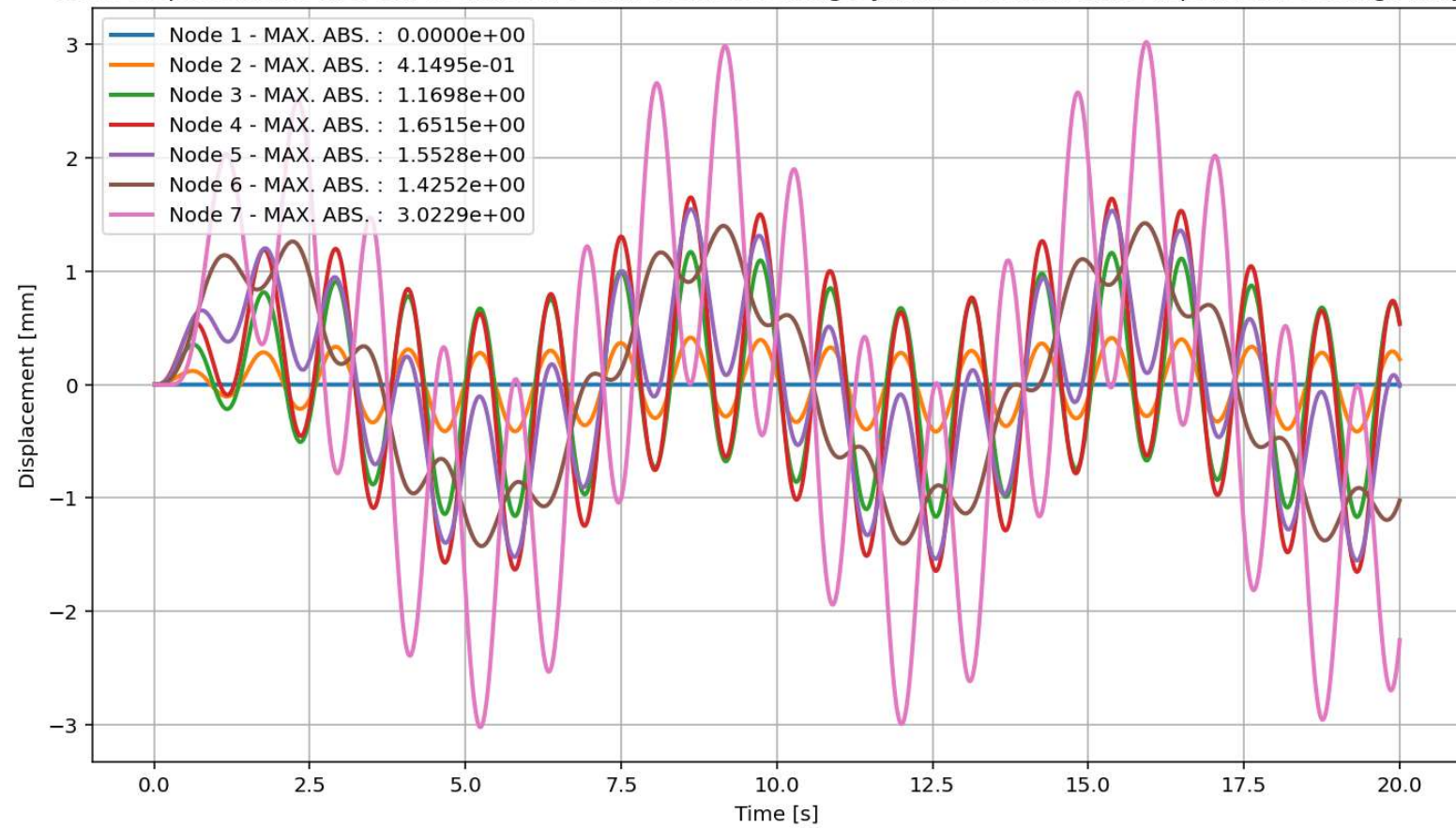
elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



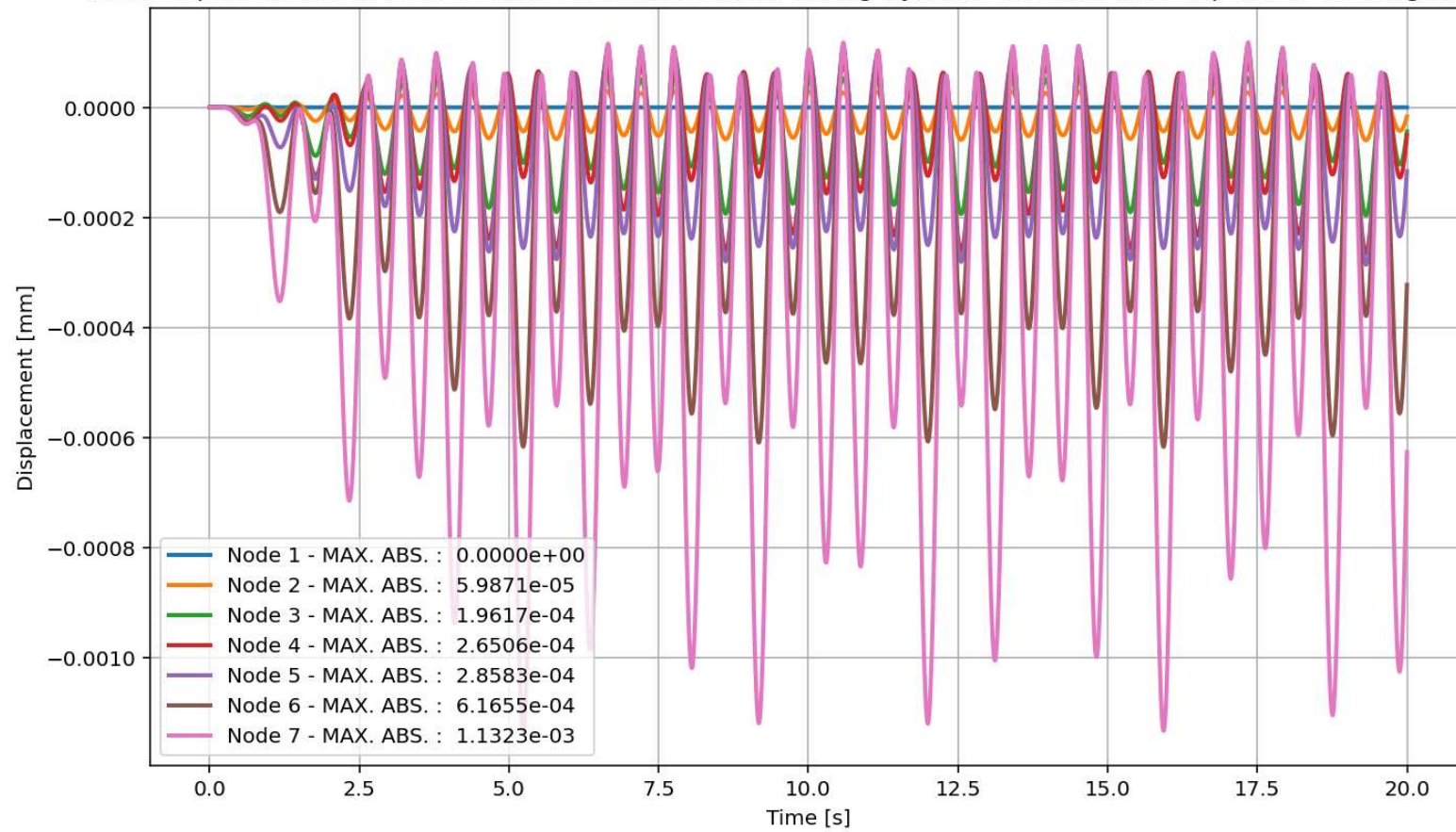
elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e8



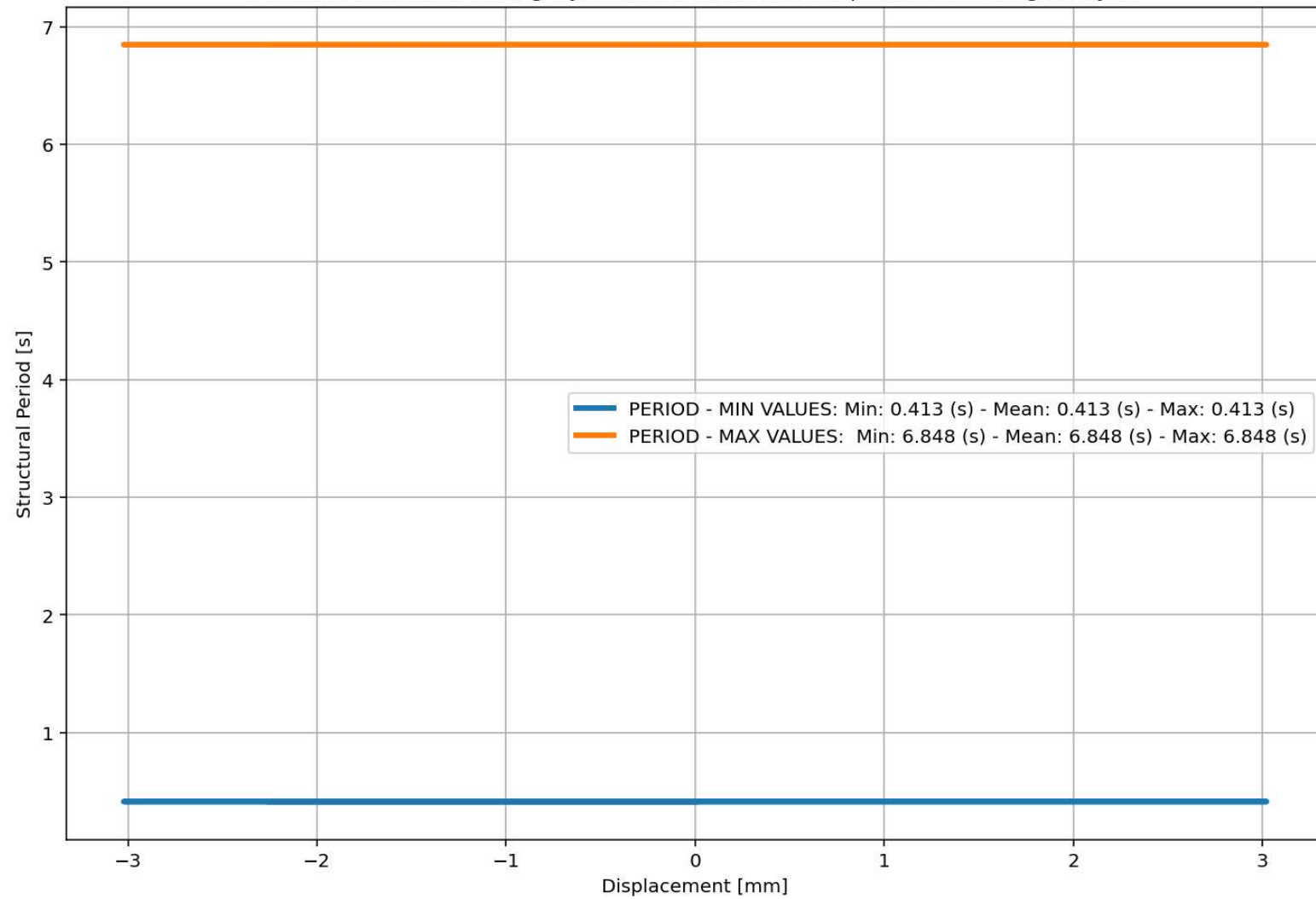
Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



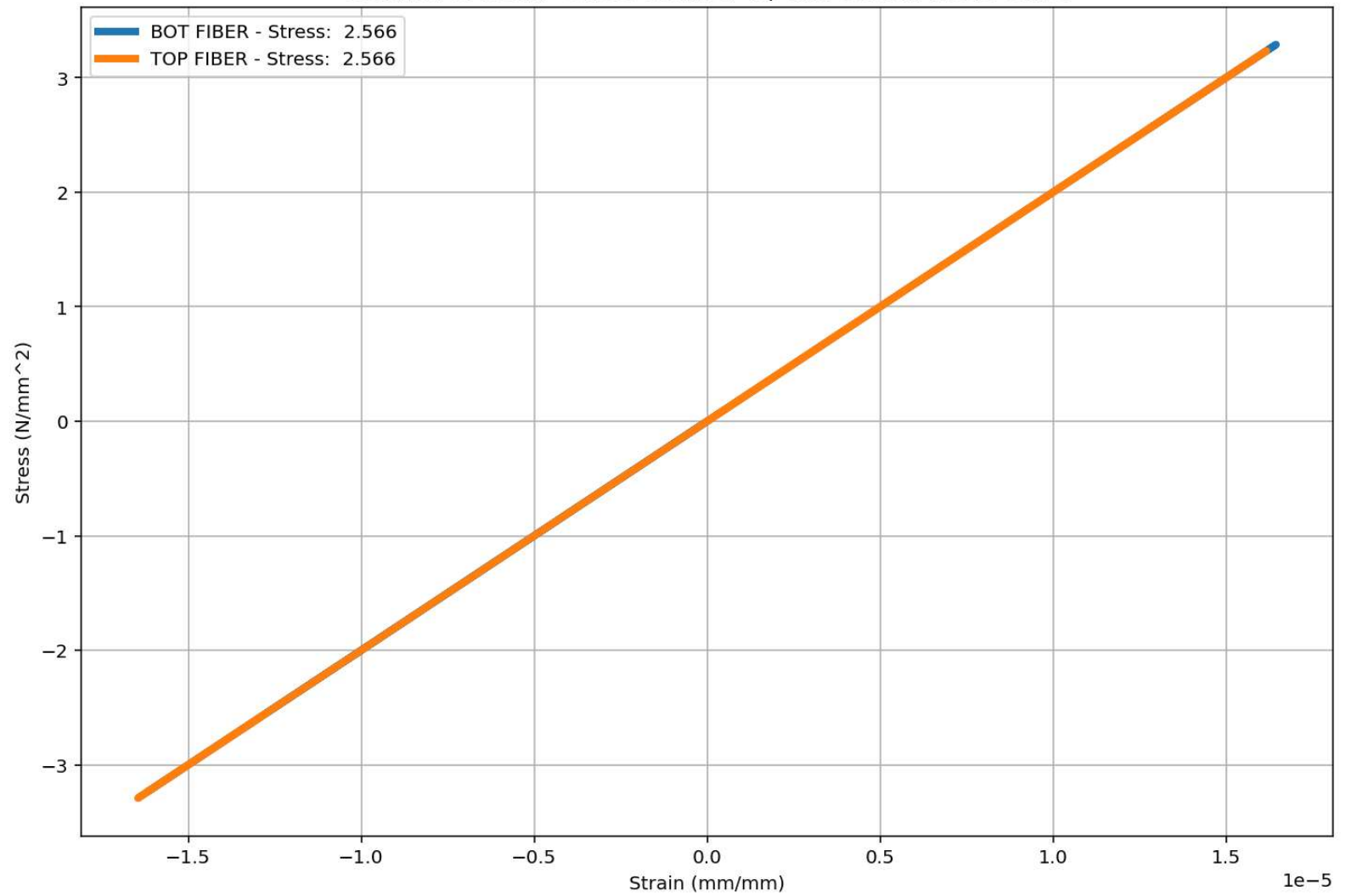
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

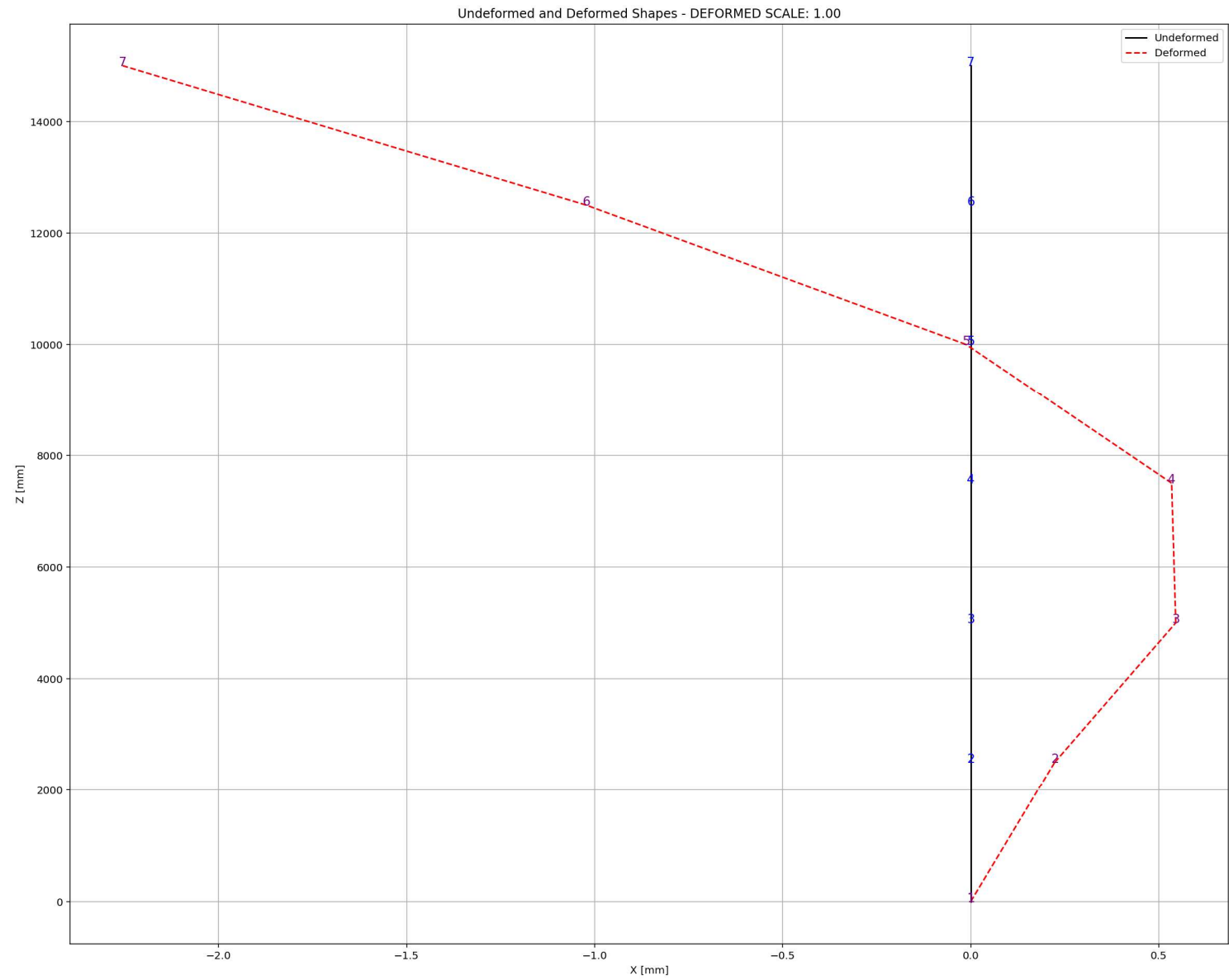


Period of Structure During Dynamic External Time-dependent Loading Analysis

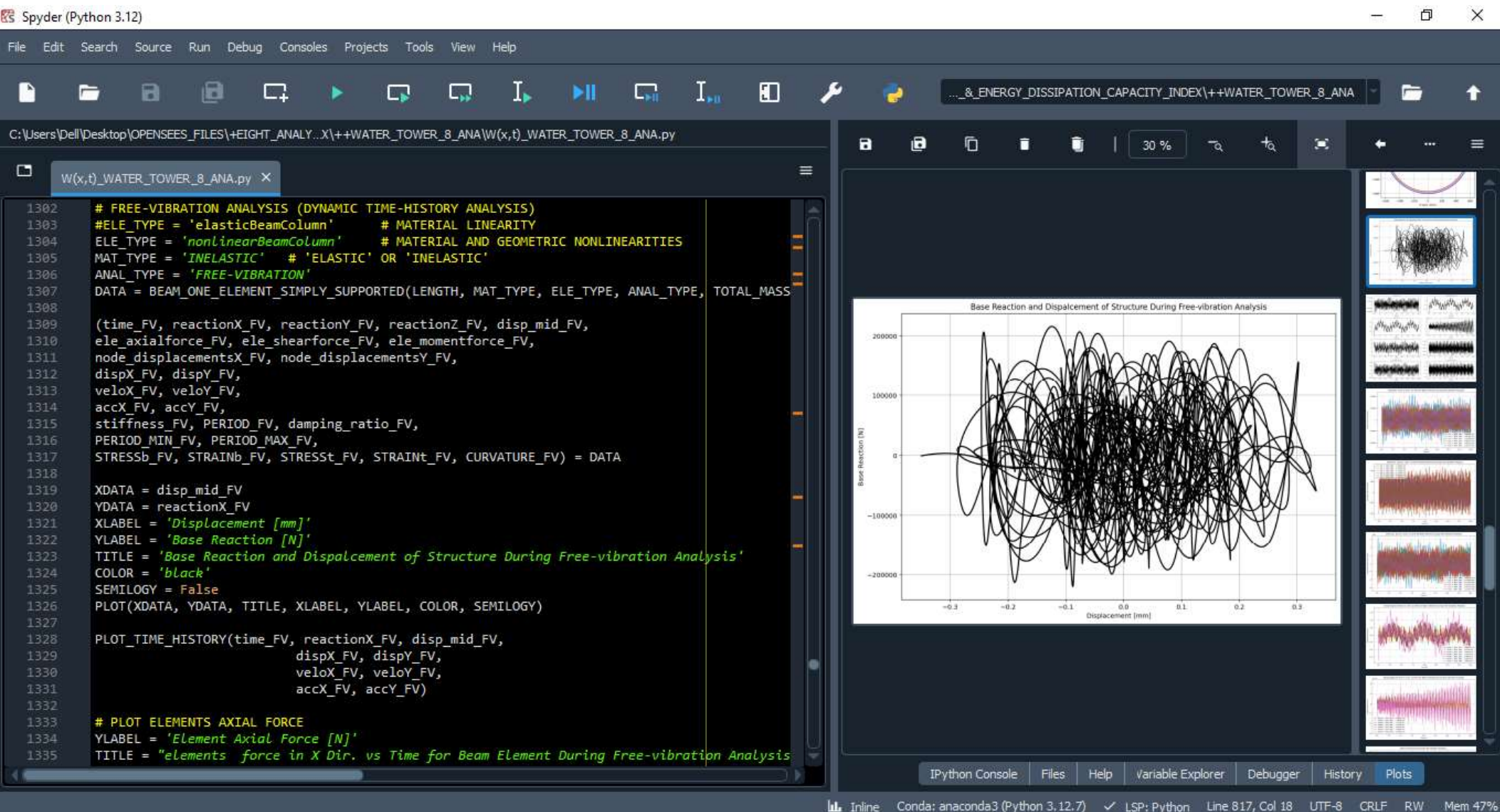


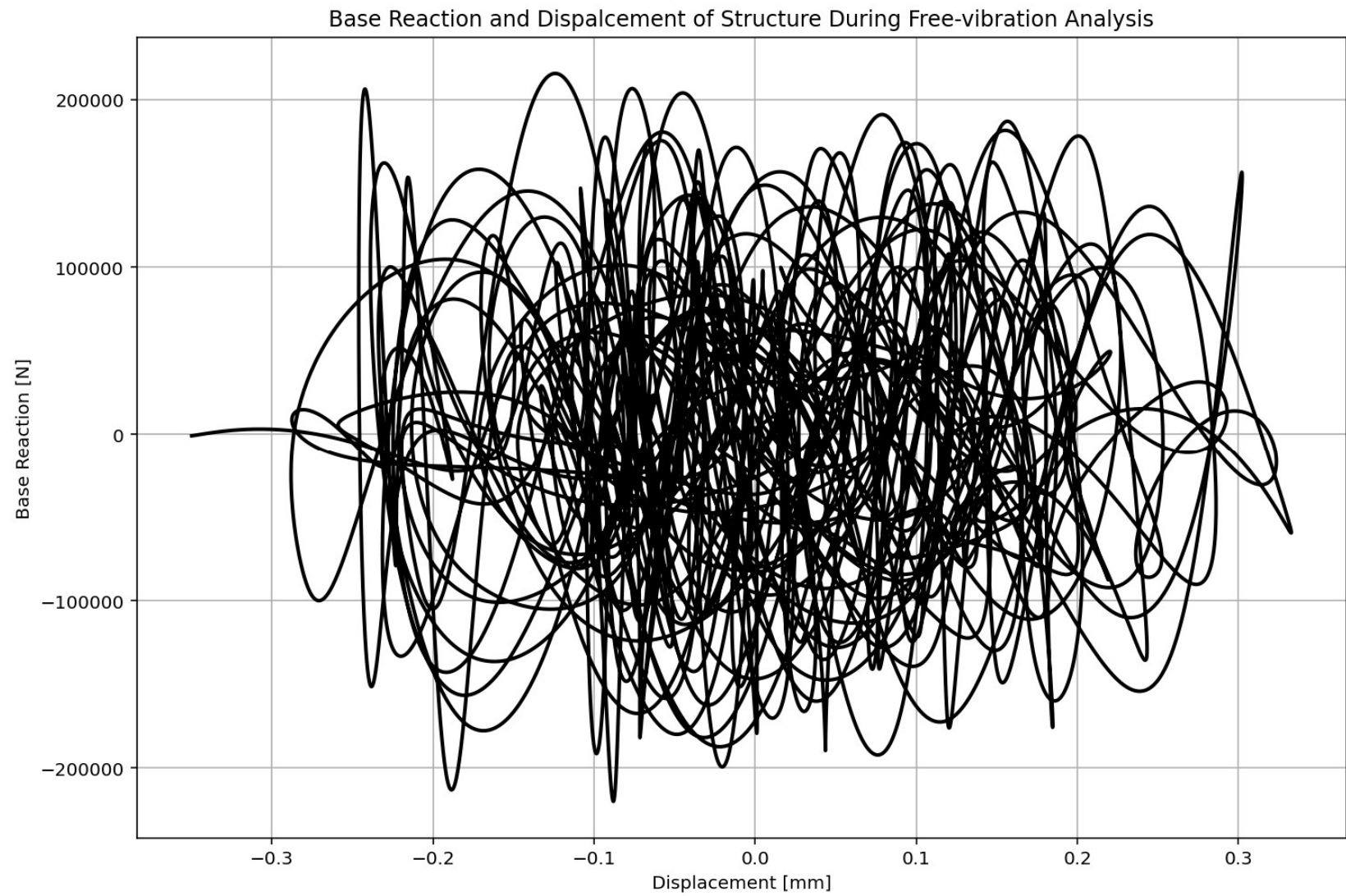
Behavior of beam - Stress-strain of Top and Bottom Fibers Curve

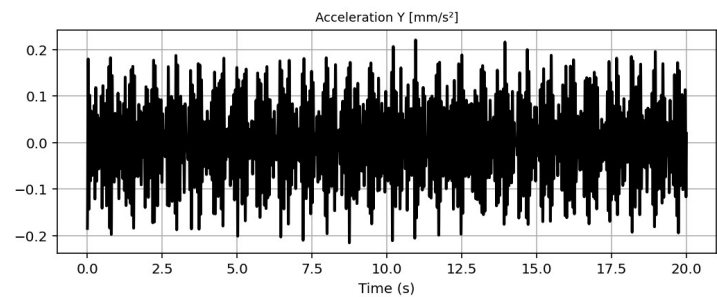
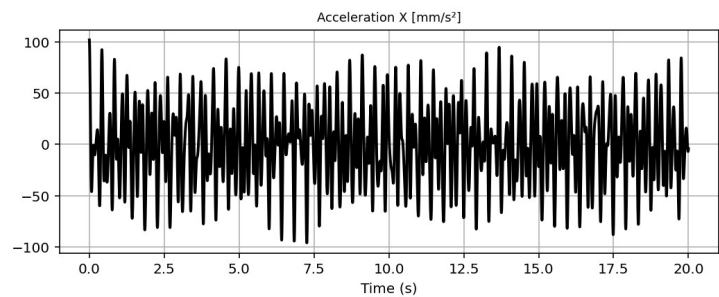
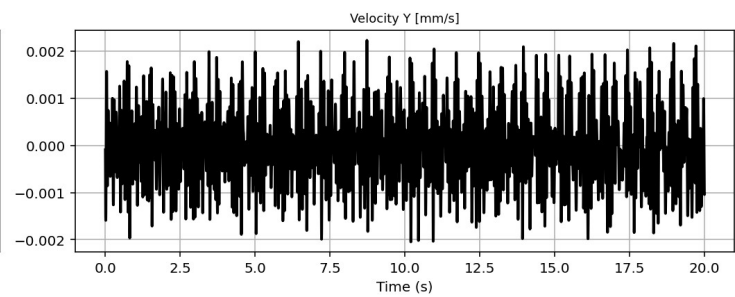
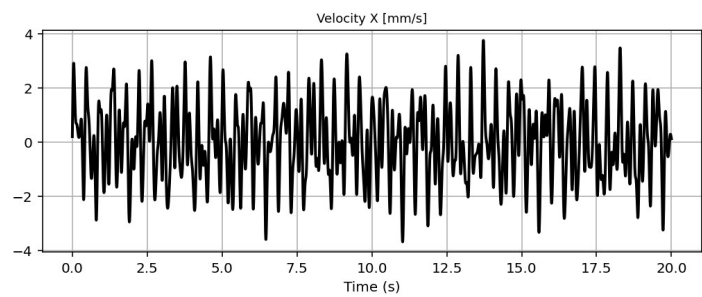
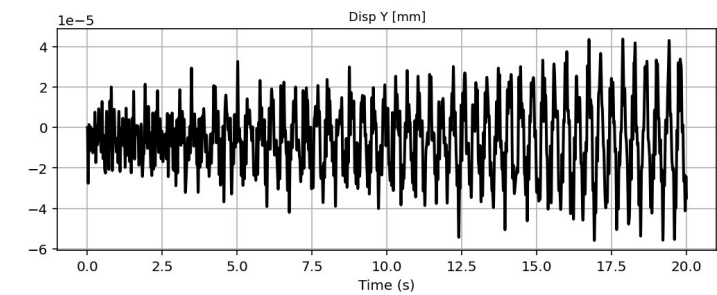
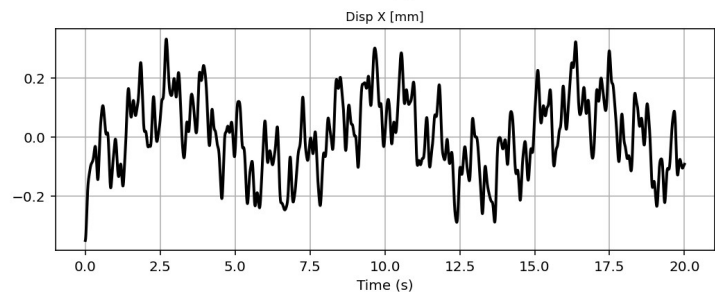
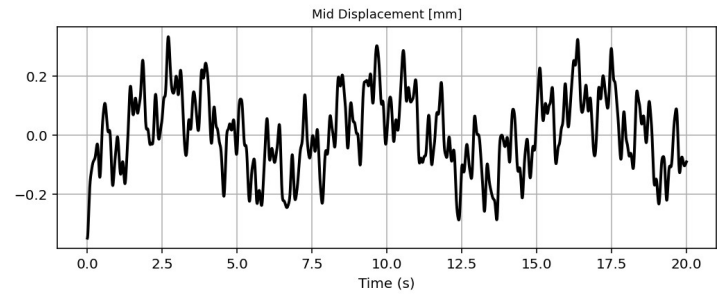
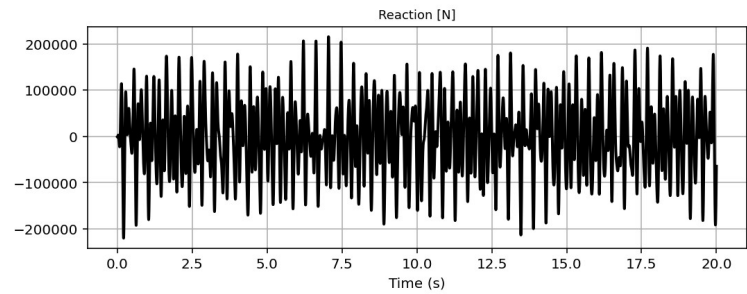




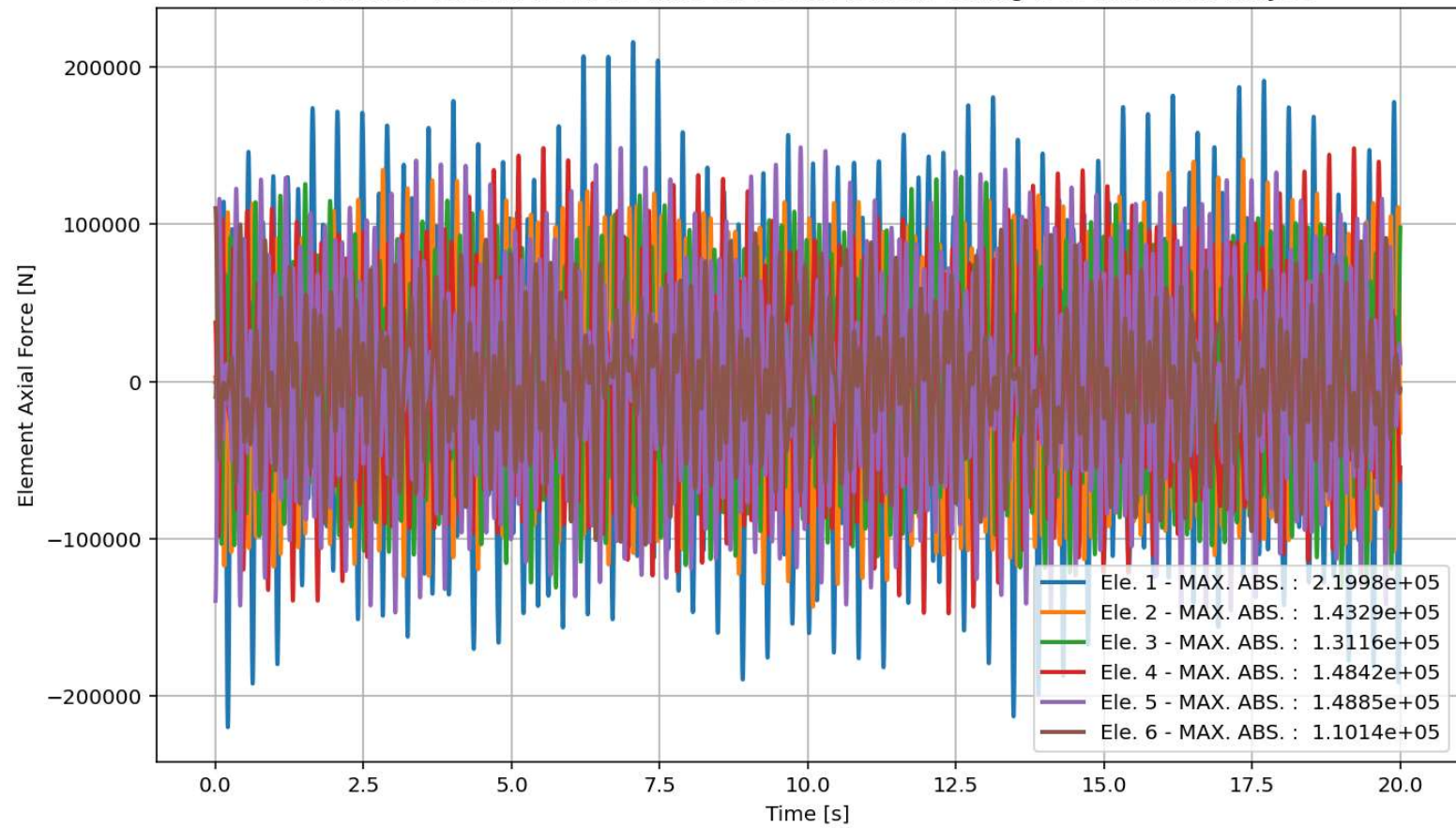
FREE-VIBRATION ANALYSIS



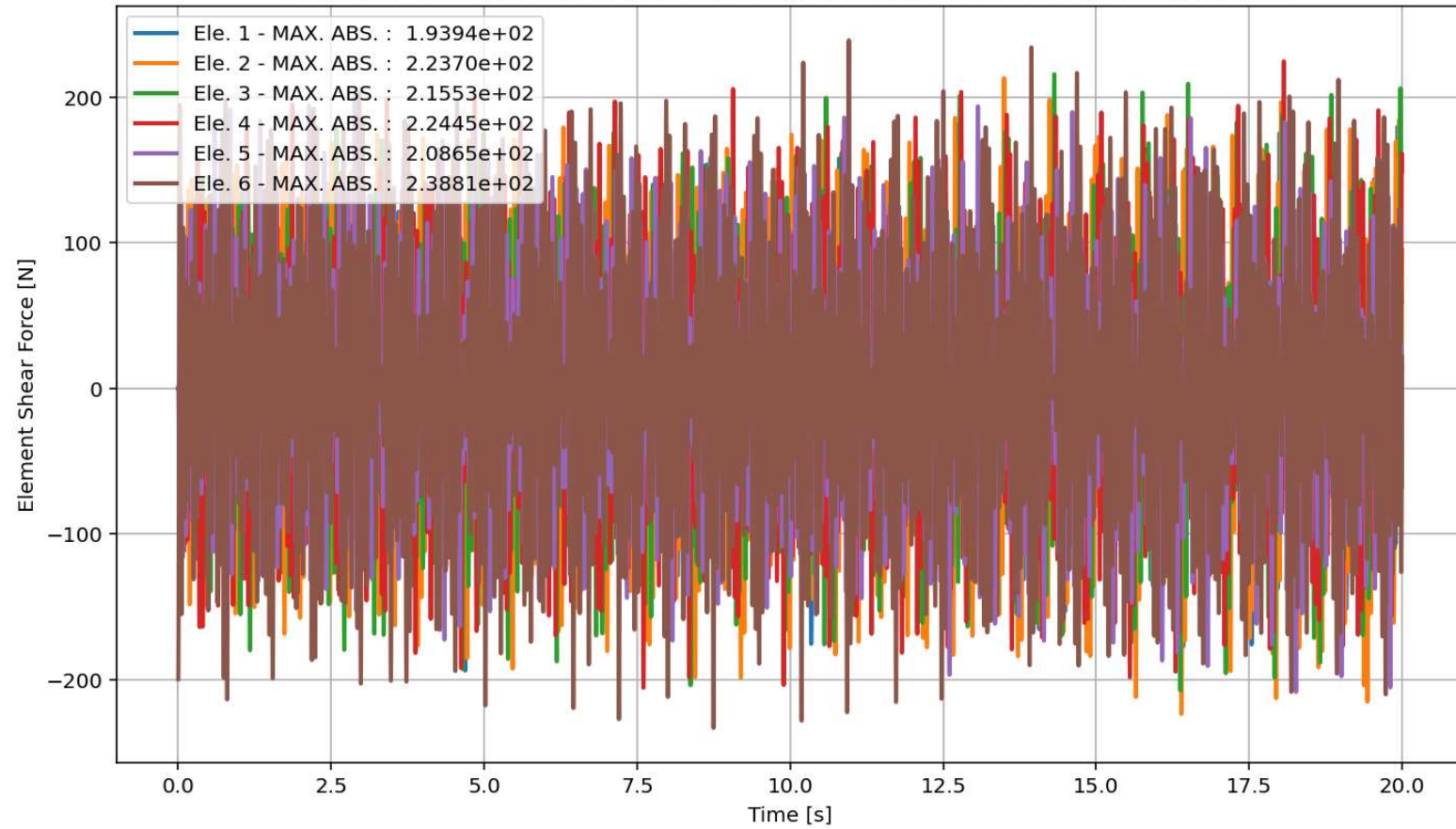


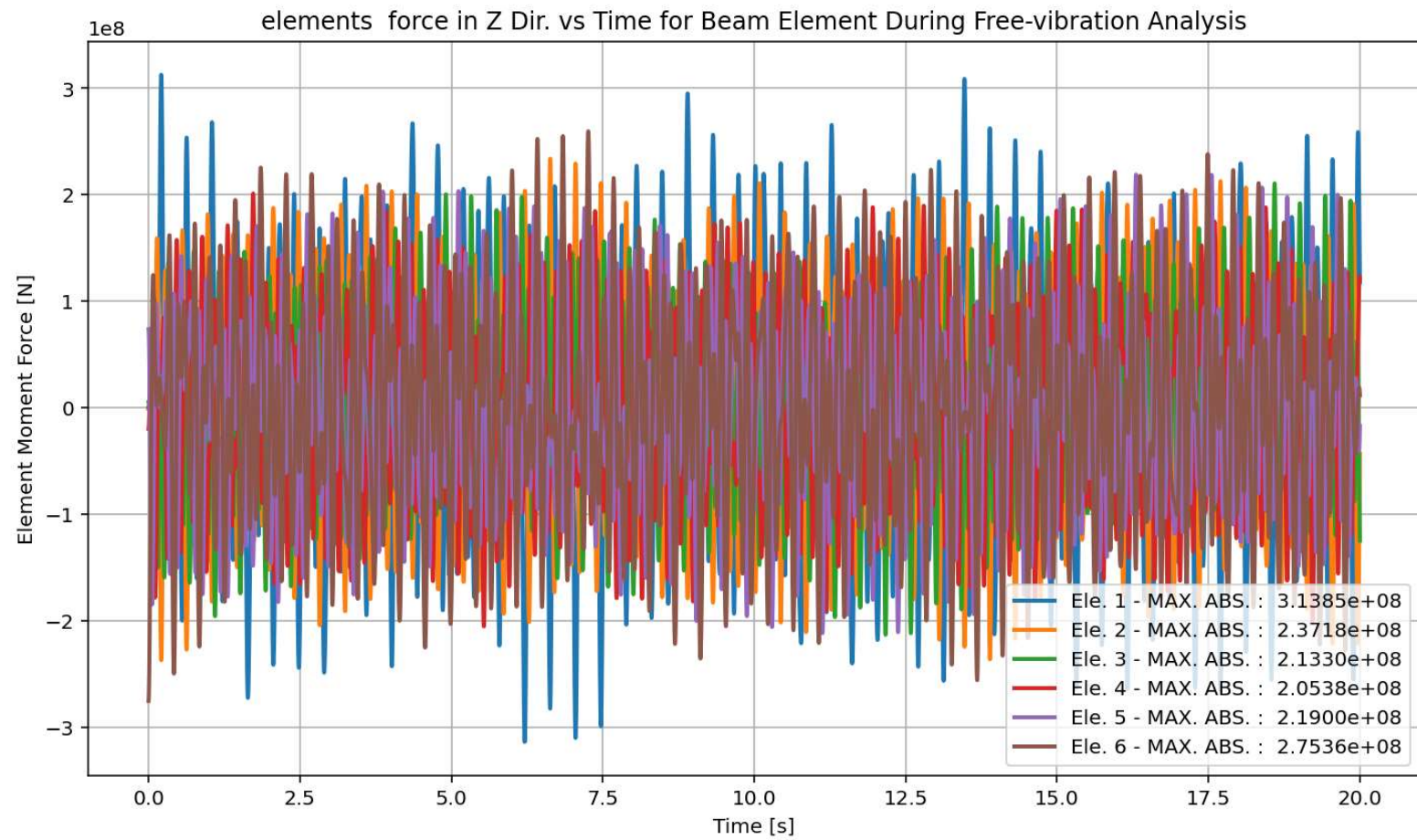


elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis

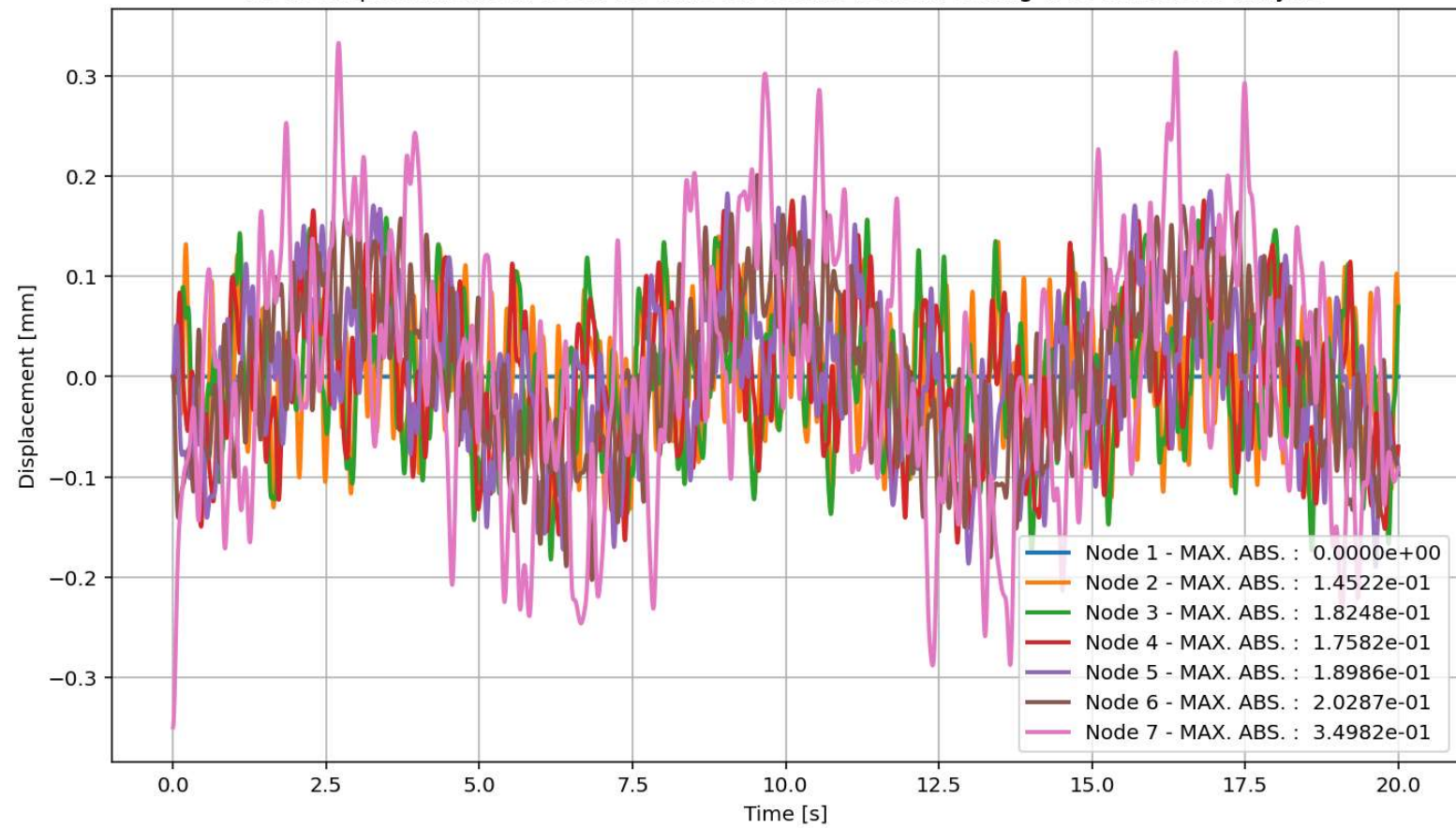


elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis

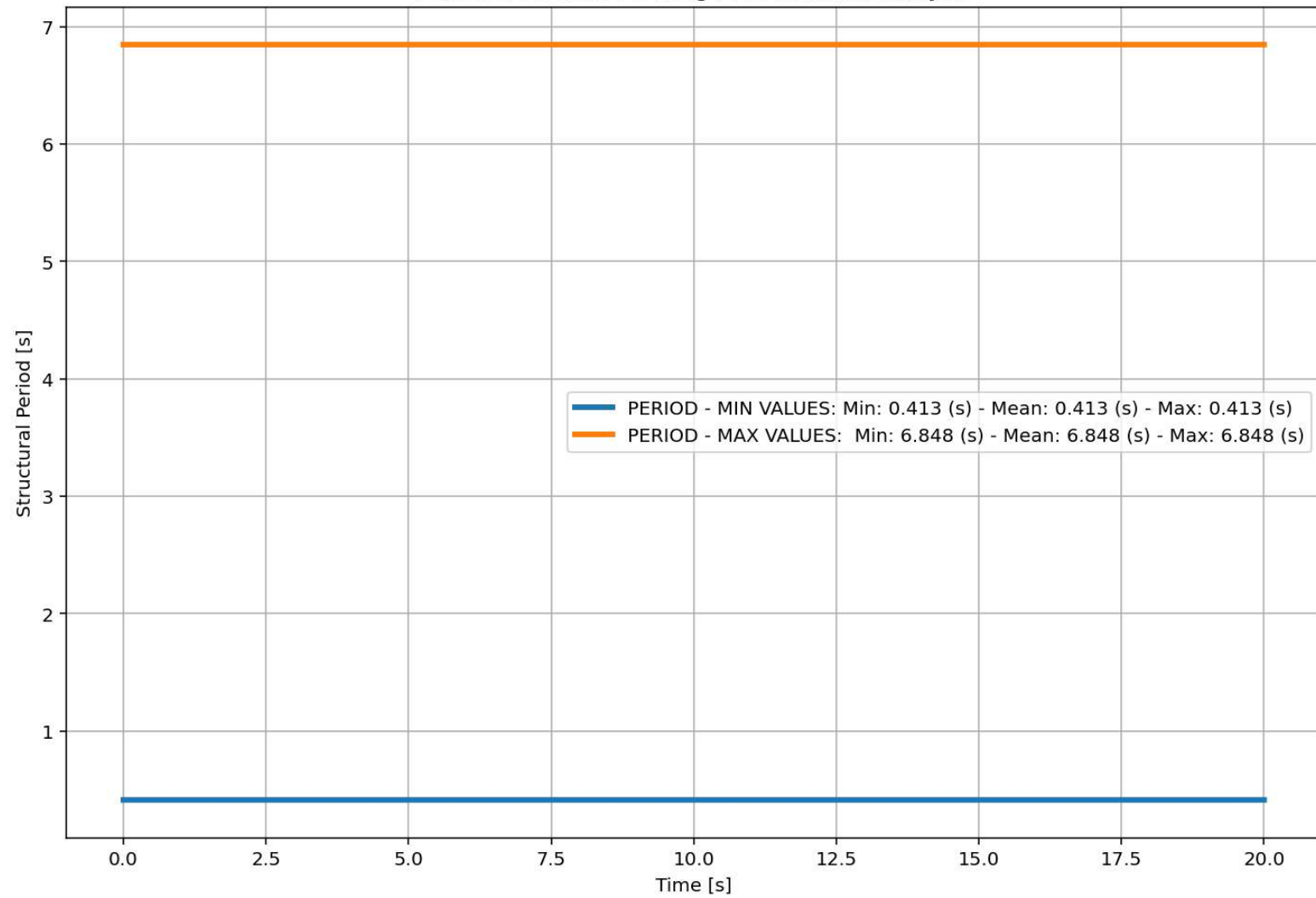




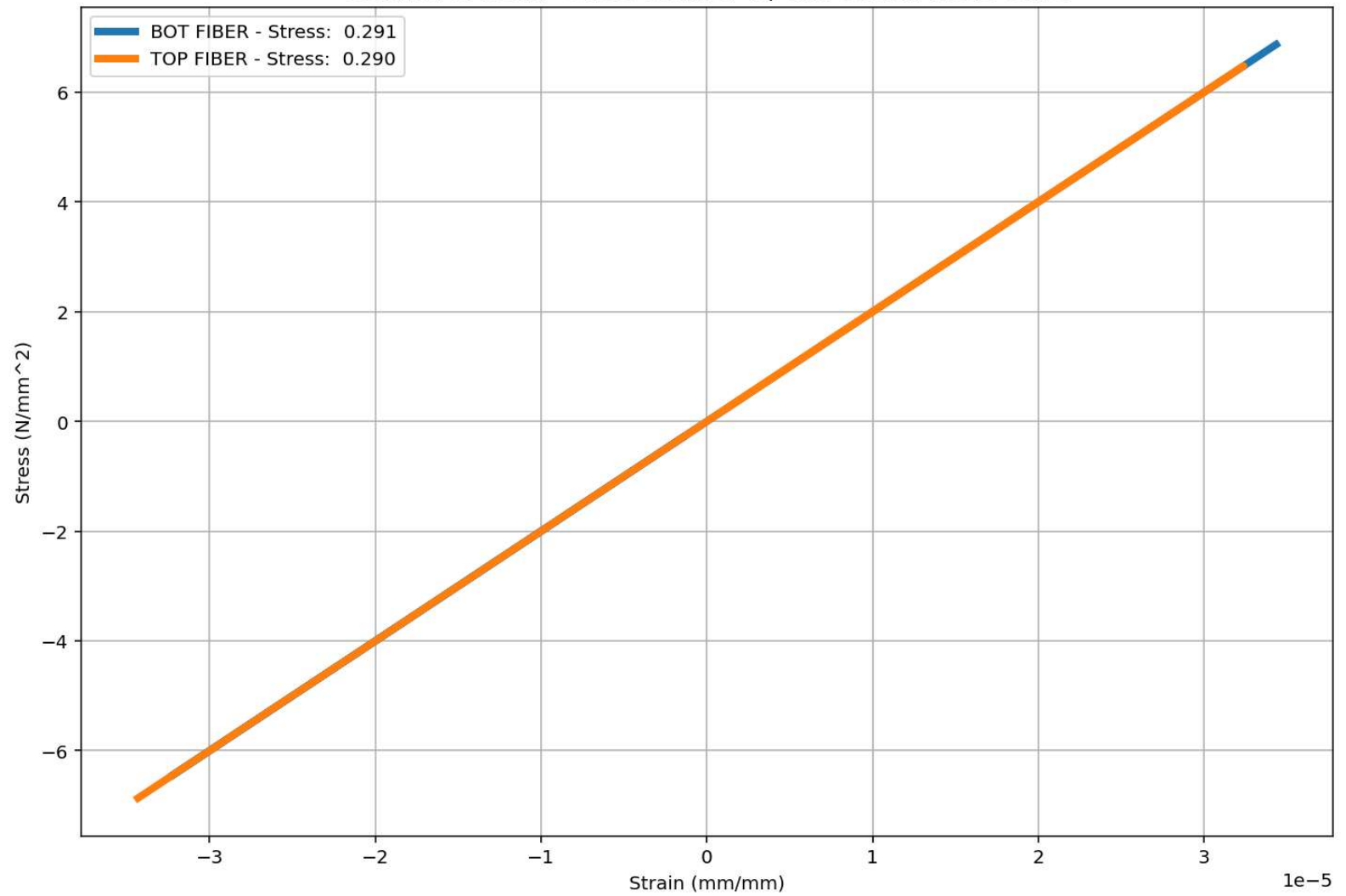
Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis



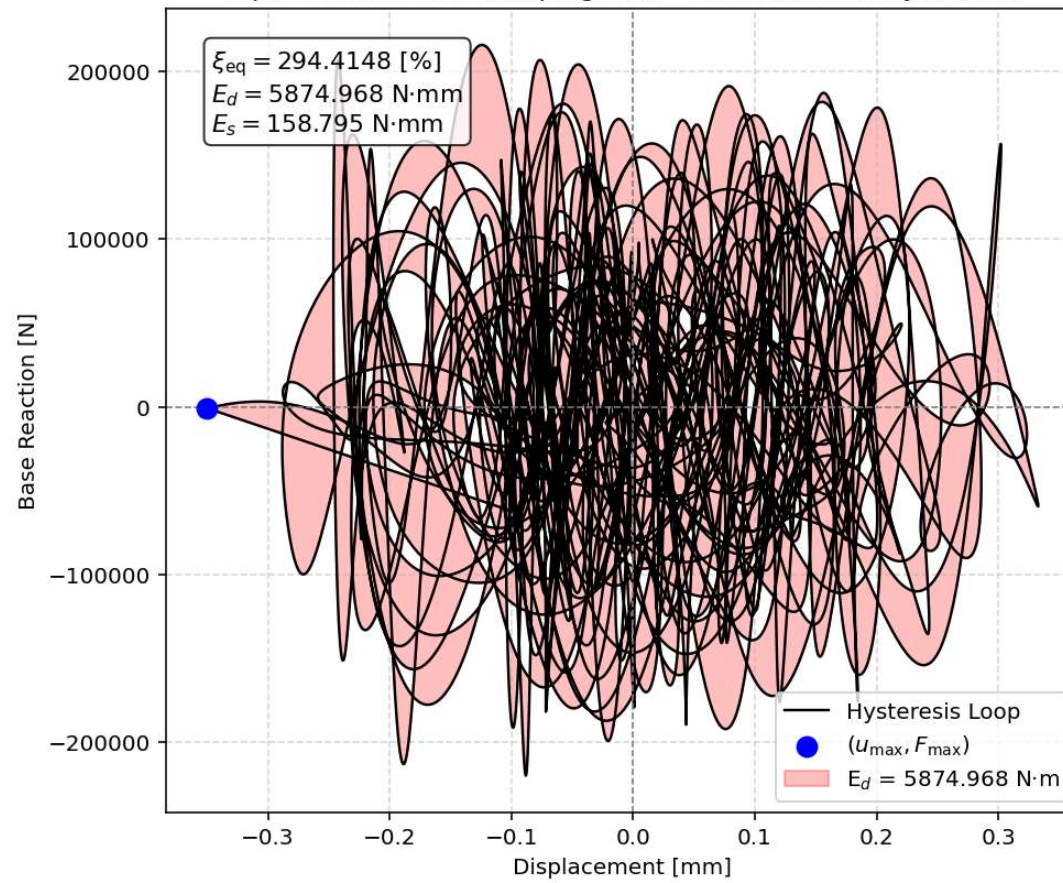
Period of Structure During Free-vibration Analysis

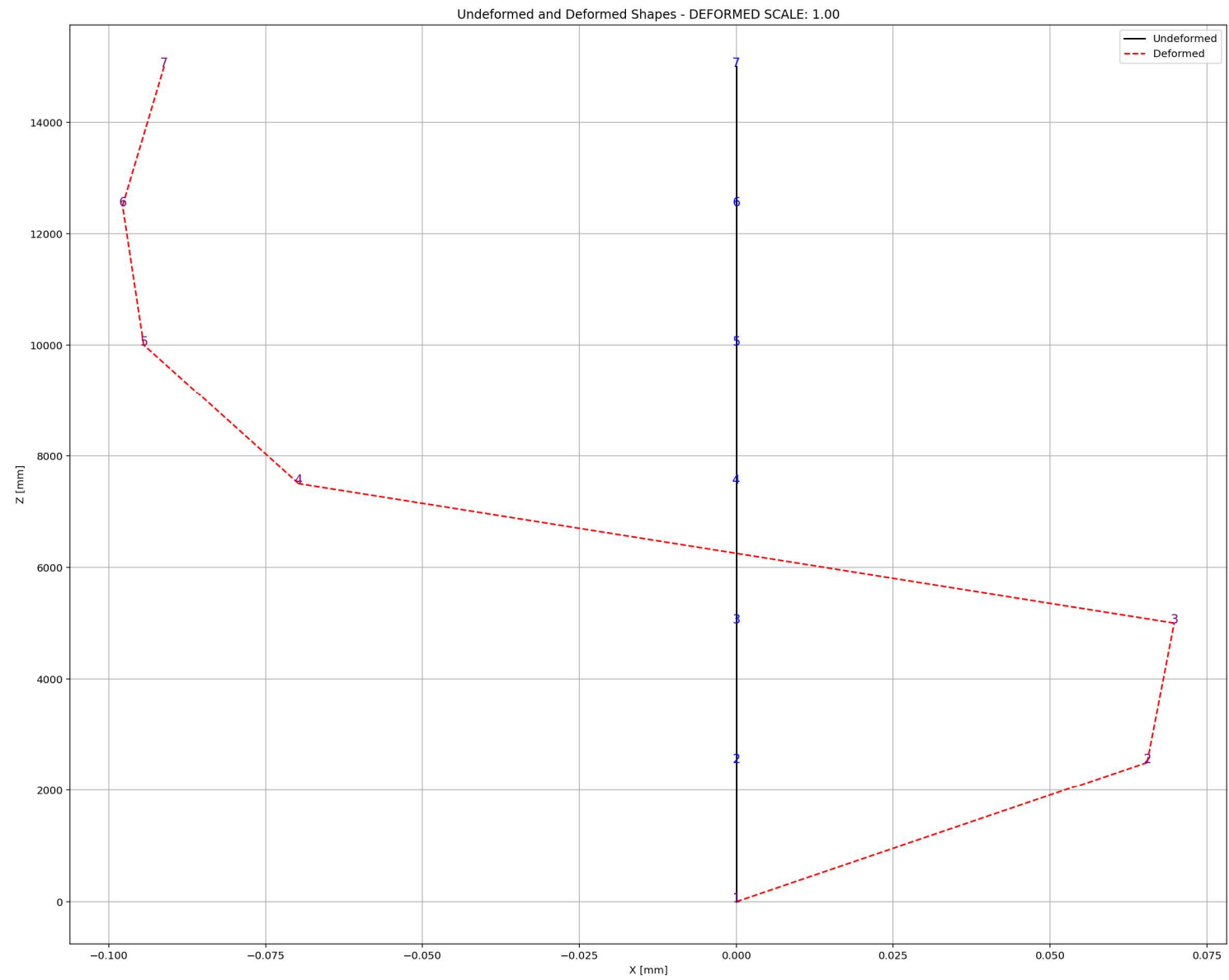


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



Equivalent viscous damping ratio - Free-vibration Hysteresis





SEISMIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...X\++WATER_TOWER_8_ANA\W(x,t)_WATER_TOWER_8_ANA.py

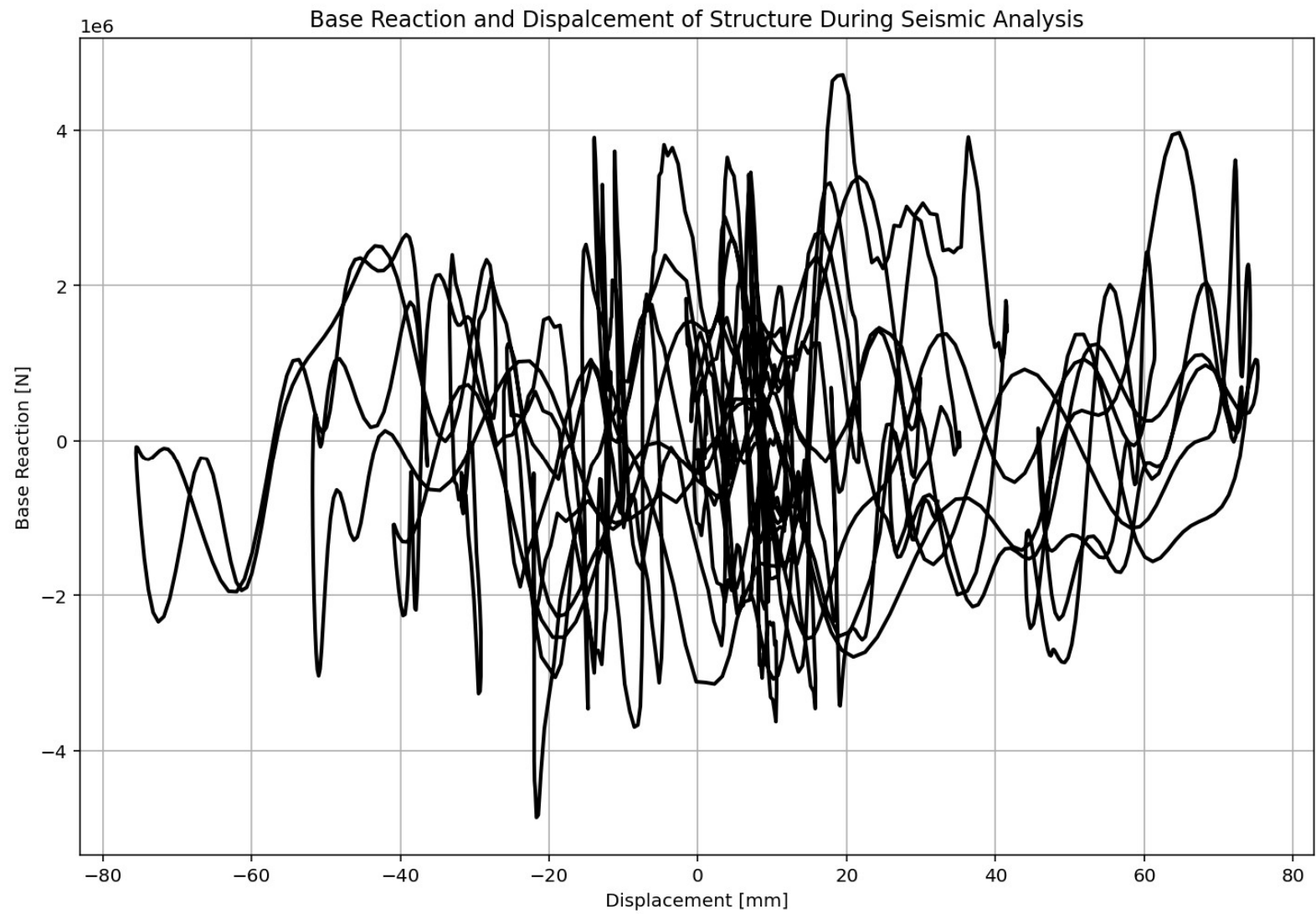
W(x,t)_WATER_TOWER_8_ANA.py X

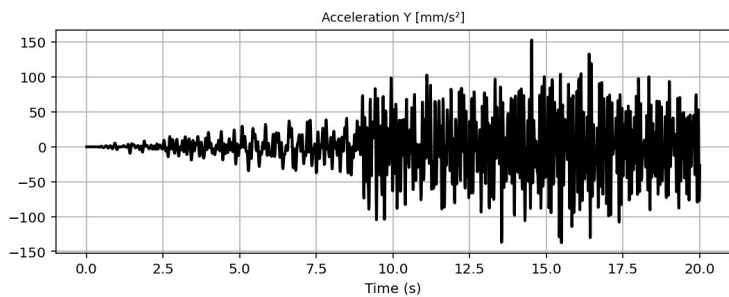
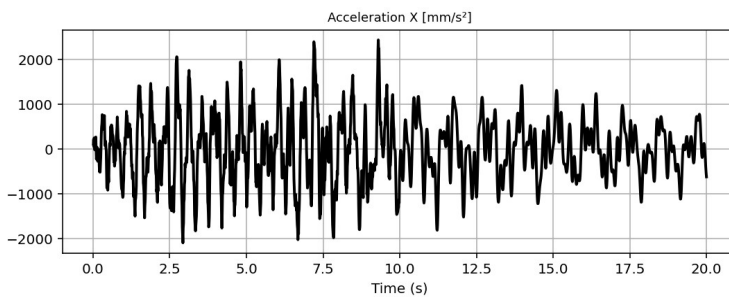
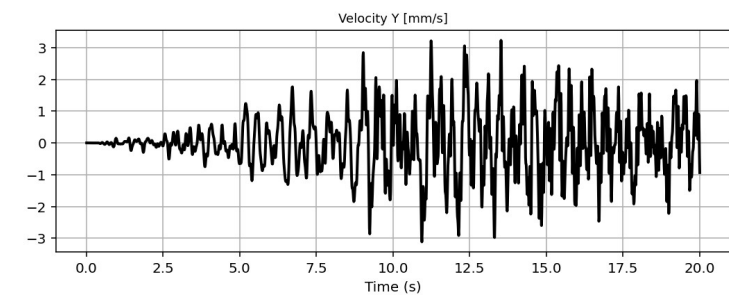
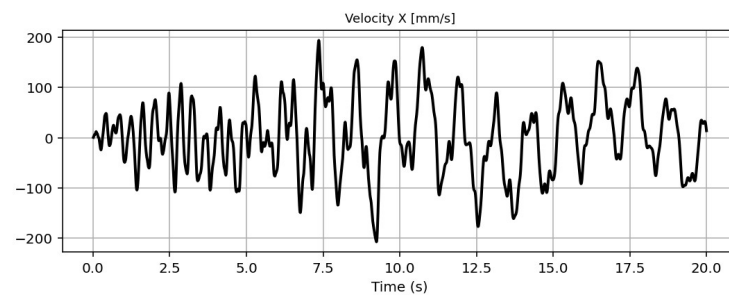
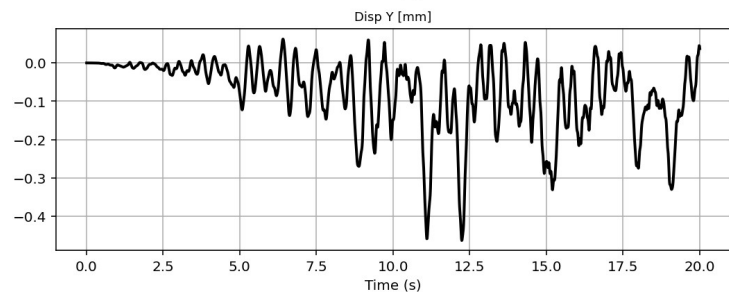
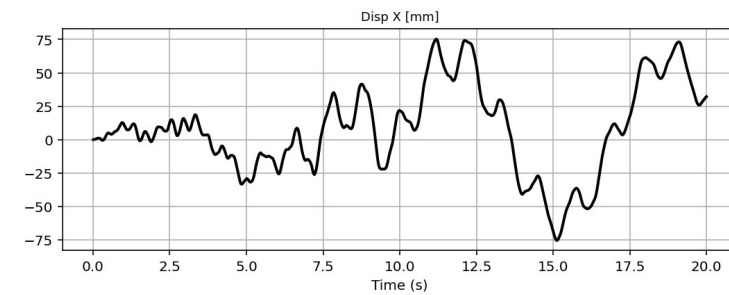
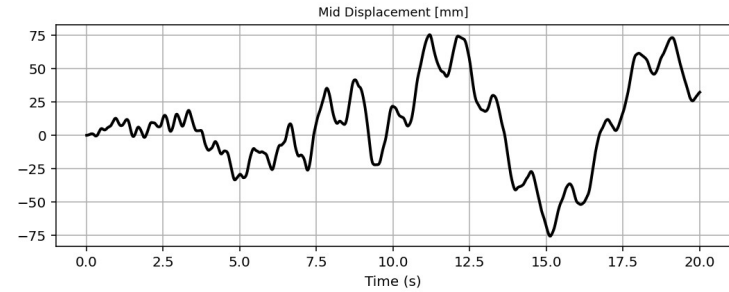
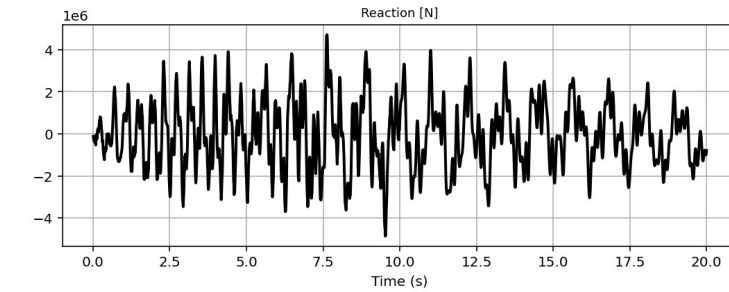
```
1377 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1378 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1379 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1380 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1381 ANAL_TYPE = 'SEISMIC'
1382
1383 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS
1384
1385 (time_SEI, reactionX_SEI, reactionY_SEI, reactionZ_SEI, disp_mid_SEI,
1386 ele_axialforce_SEI, ele_shearforce_SEI, ele_momentforce_SEI,
1387 node_displacementsX_SEI, node_displacementsY_SEI,
1388 dispX_SEI, dispY_SEI,
1389 veloX_SEI, veloY_SEI,
1390 accX_SEI, accY_SEI,
1391 stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1392 PERIOD_MIN_SEI, PERIOD_MAX_SEI,
1393 STRESSb_SEI, STRAINb_SEI, STRESSt_SEI, STRAINt_SEI, CURVATURE_SEI) = DATA
1394
1395
1396 XDATA = disp_mid_SEI
1397 YDATA = reactionX_SEI
1398 XLABEL = 'Displacement [mm]'
1399 YLABEL = 'Base Reaction [N]'
1400 TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
1401 COLOR = 'black'
1402 SEMILOGY = False
1403 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1404
1405 PLOT_TIME_HISTORY(time_SEI, reactionX_SEI, disp_mid_SEI,
1406 dispX_SEI, dispY_SEI,
1407 veloX_SEI, veloY_SEI,
1408 accX_SEI, accY_SEI)
1409
1410 # PLOT ELEMENTS AXIAL FORCE
```

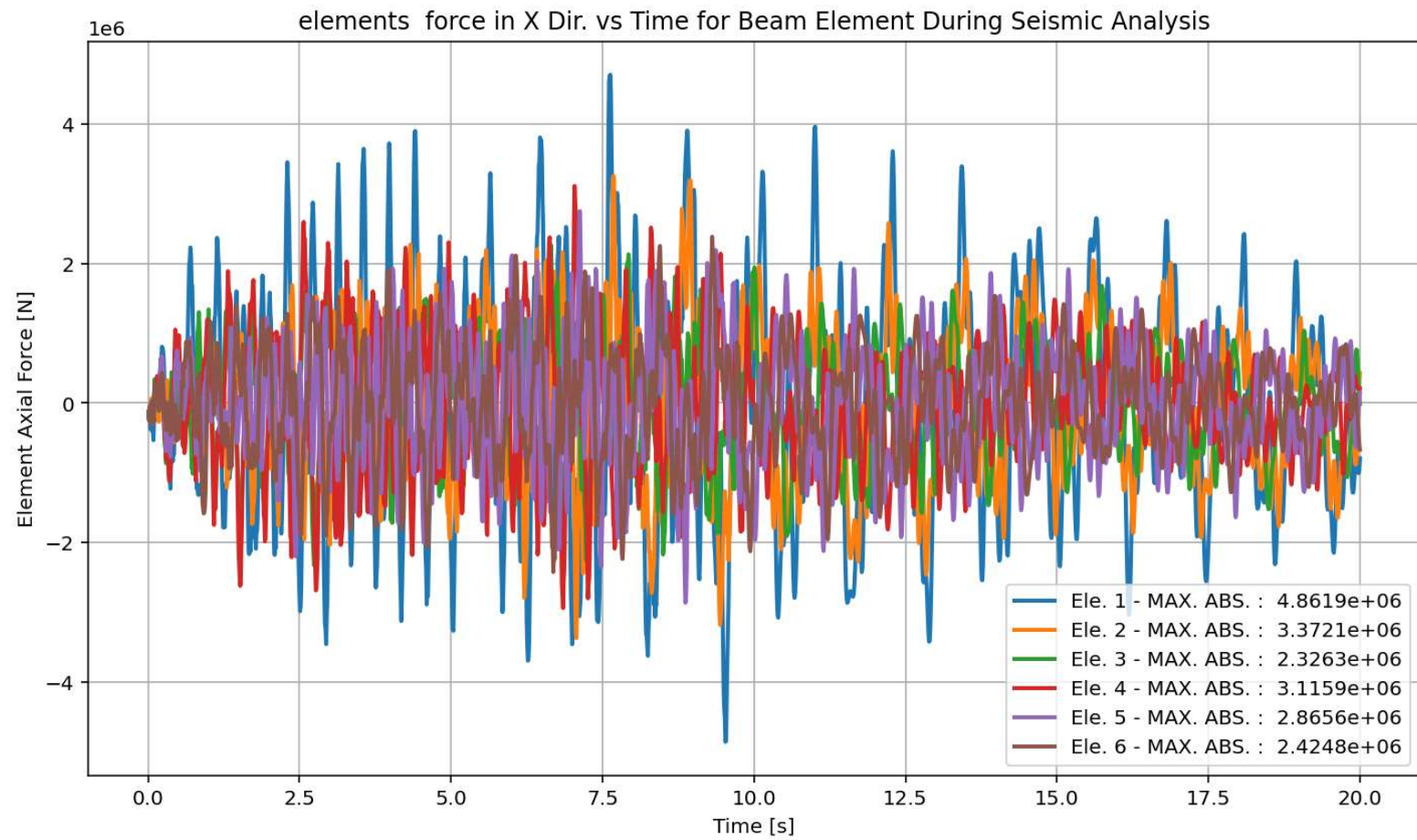
Base Reaction and Dispalcement of Structure During Seismic Analysis

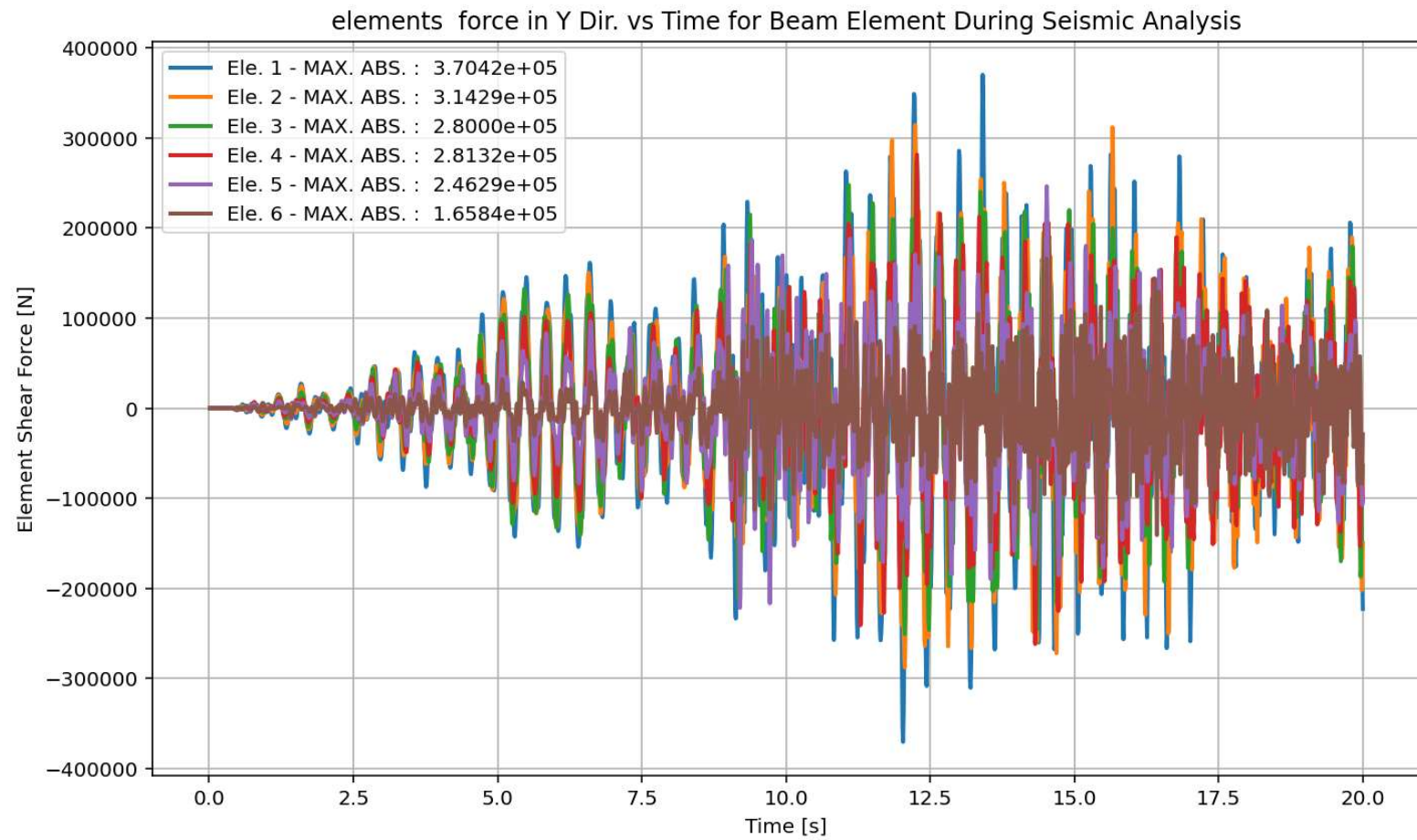
IPython Console Files Help Variable Explorer Debugger History Plots

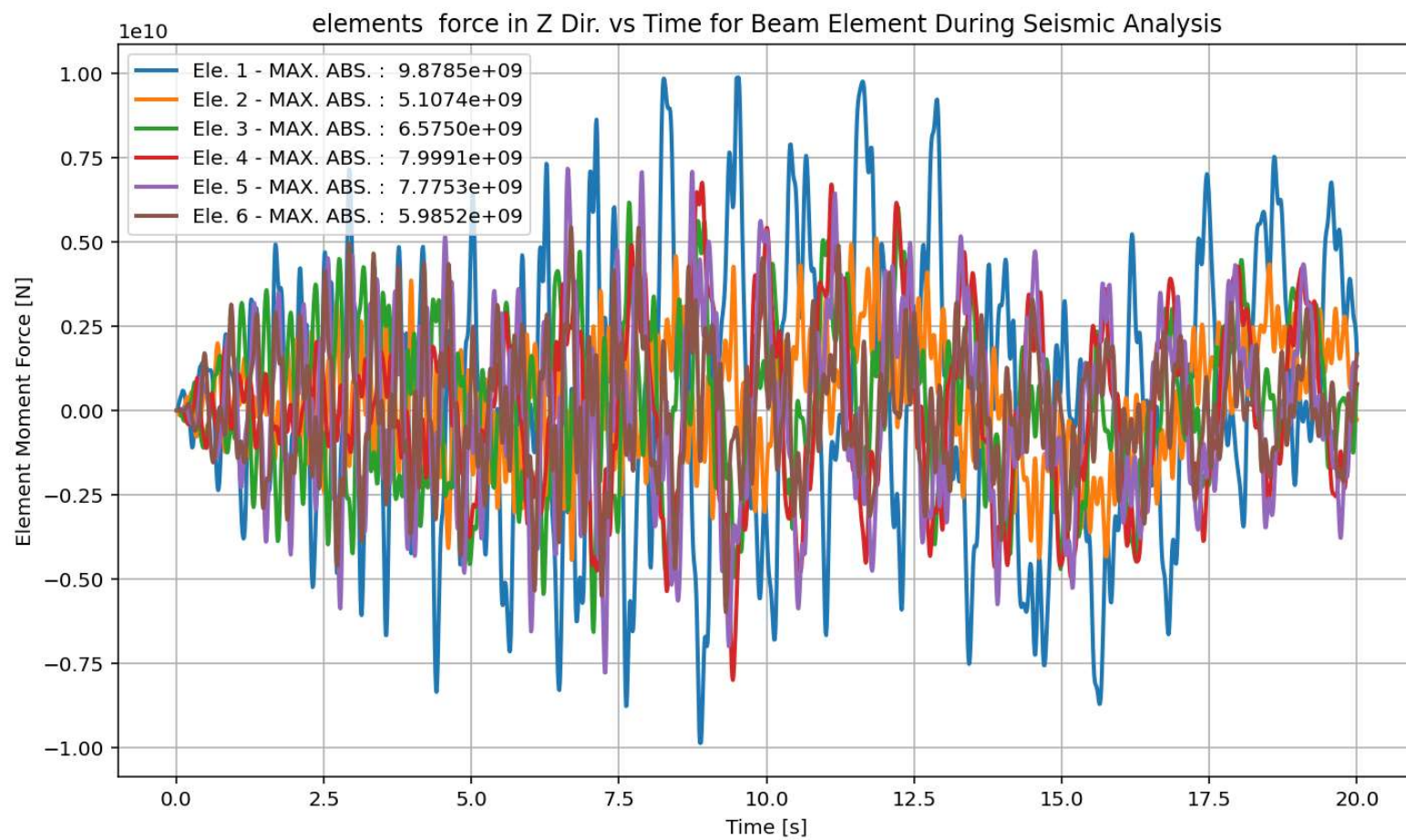
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 817, Col 18 UTF-8 CRLF RW Mem 46%



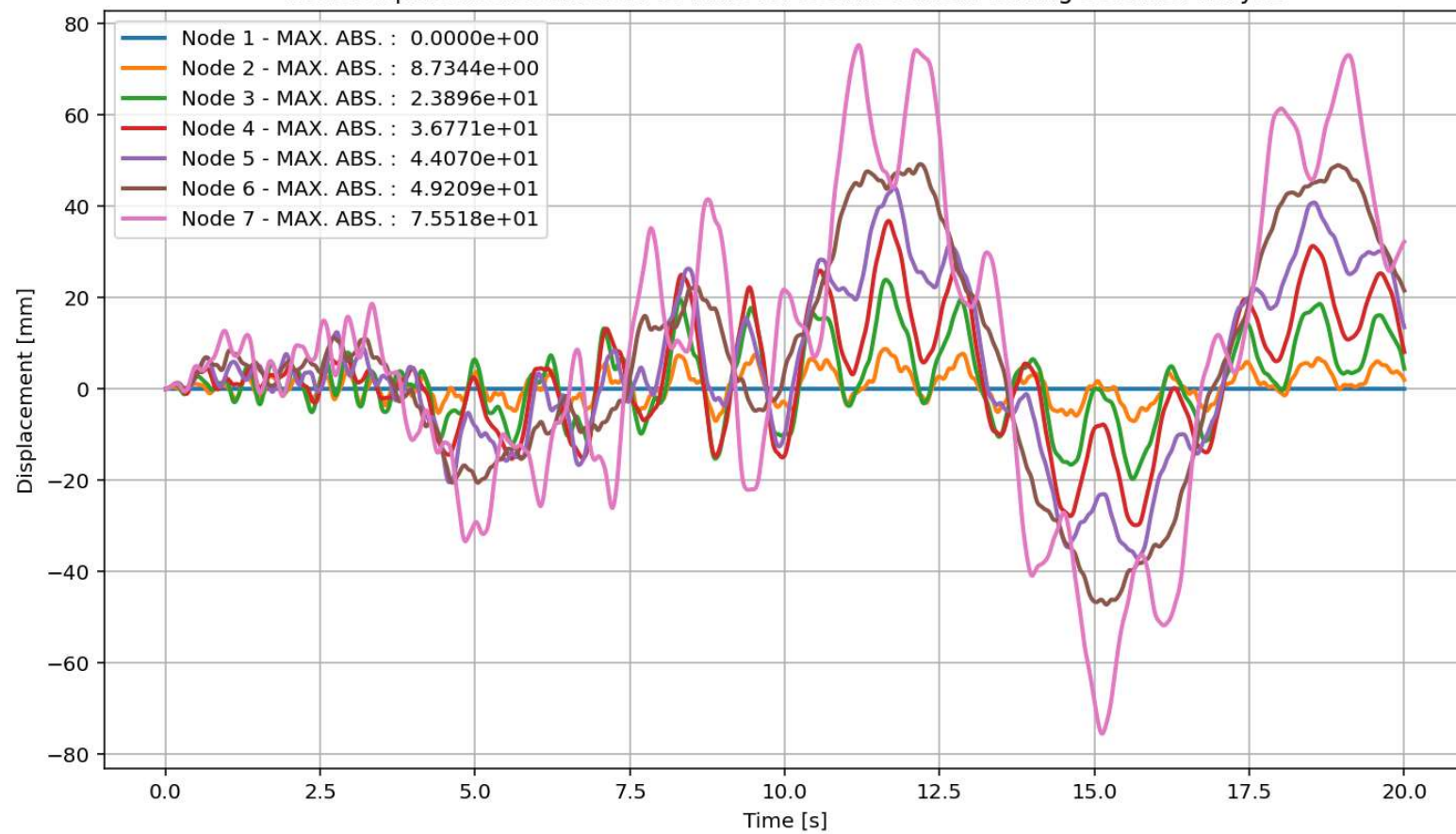


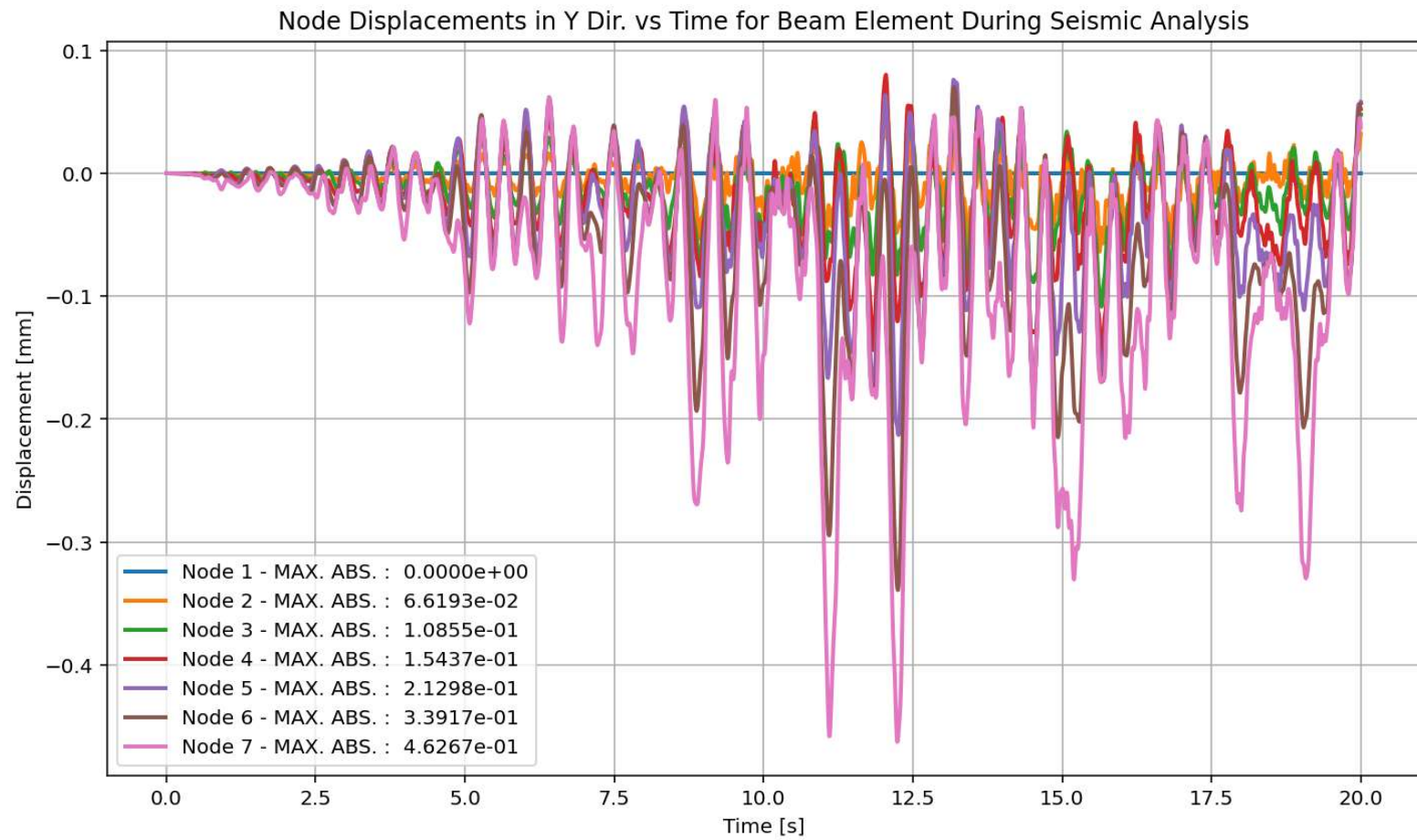




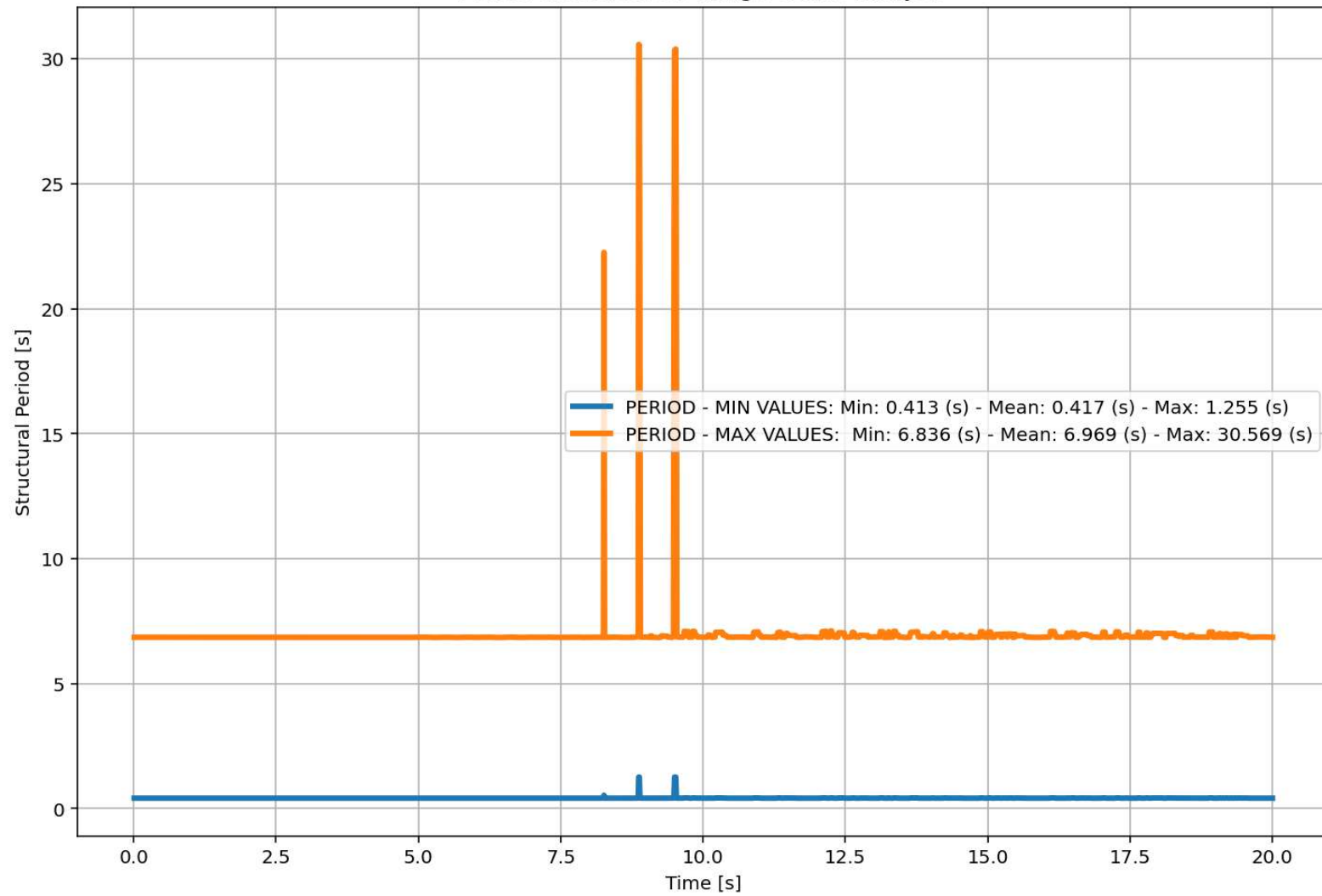


Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis

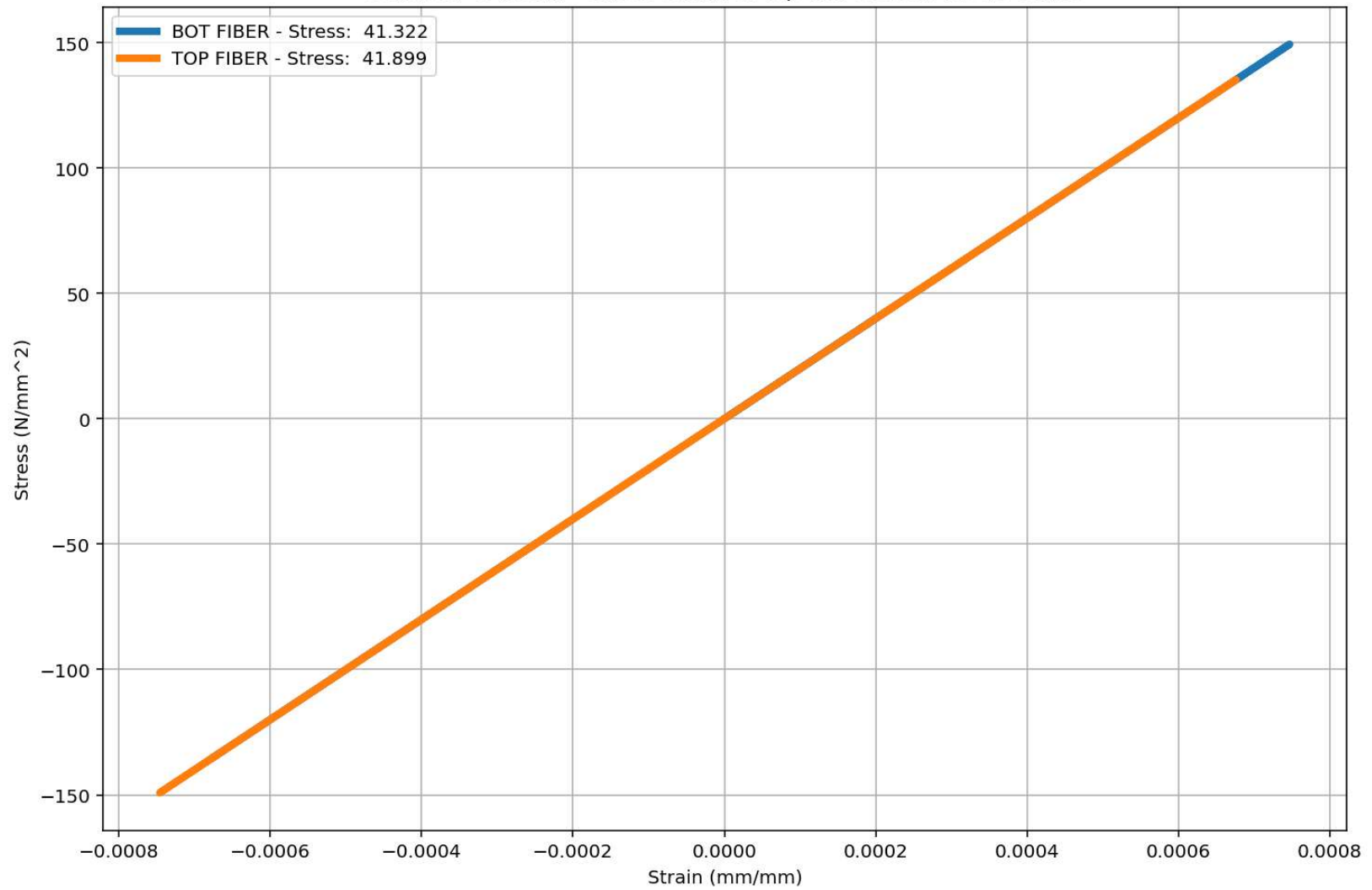


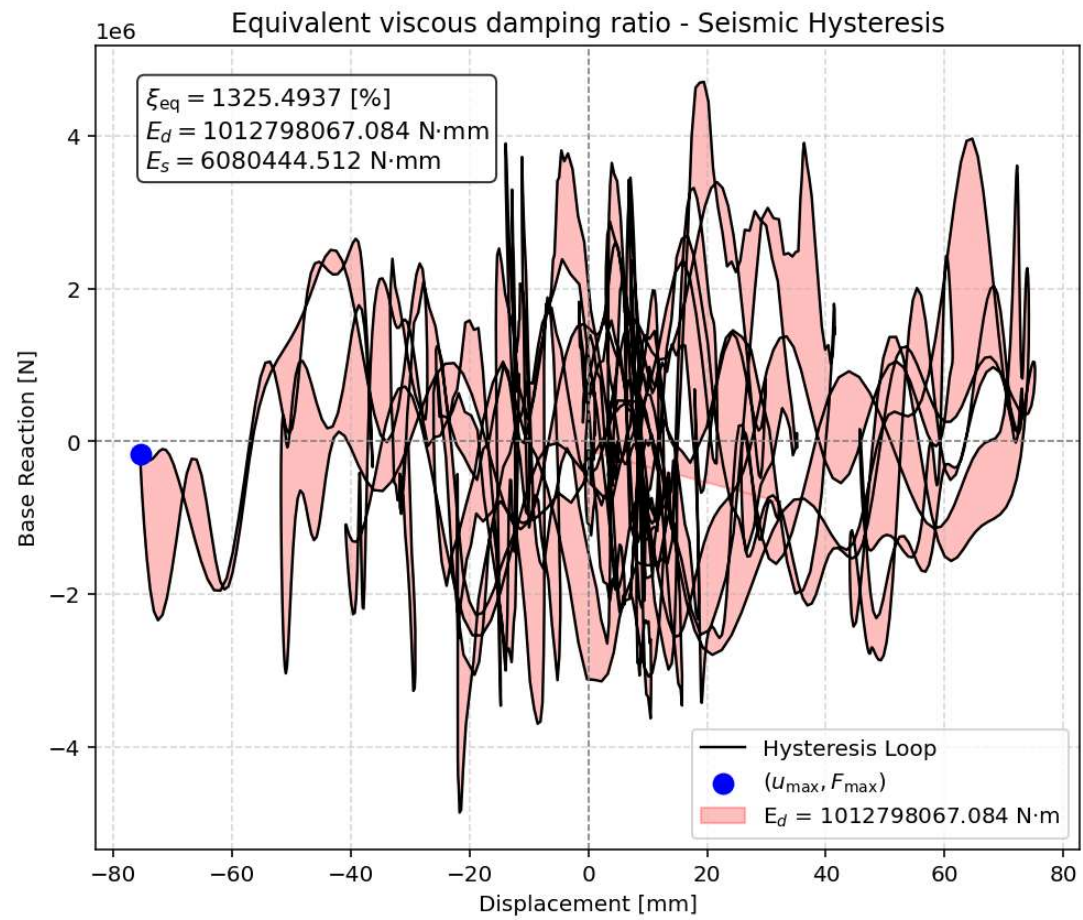


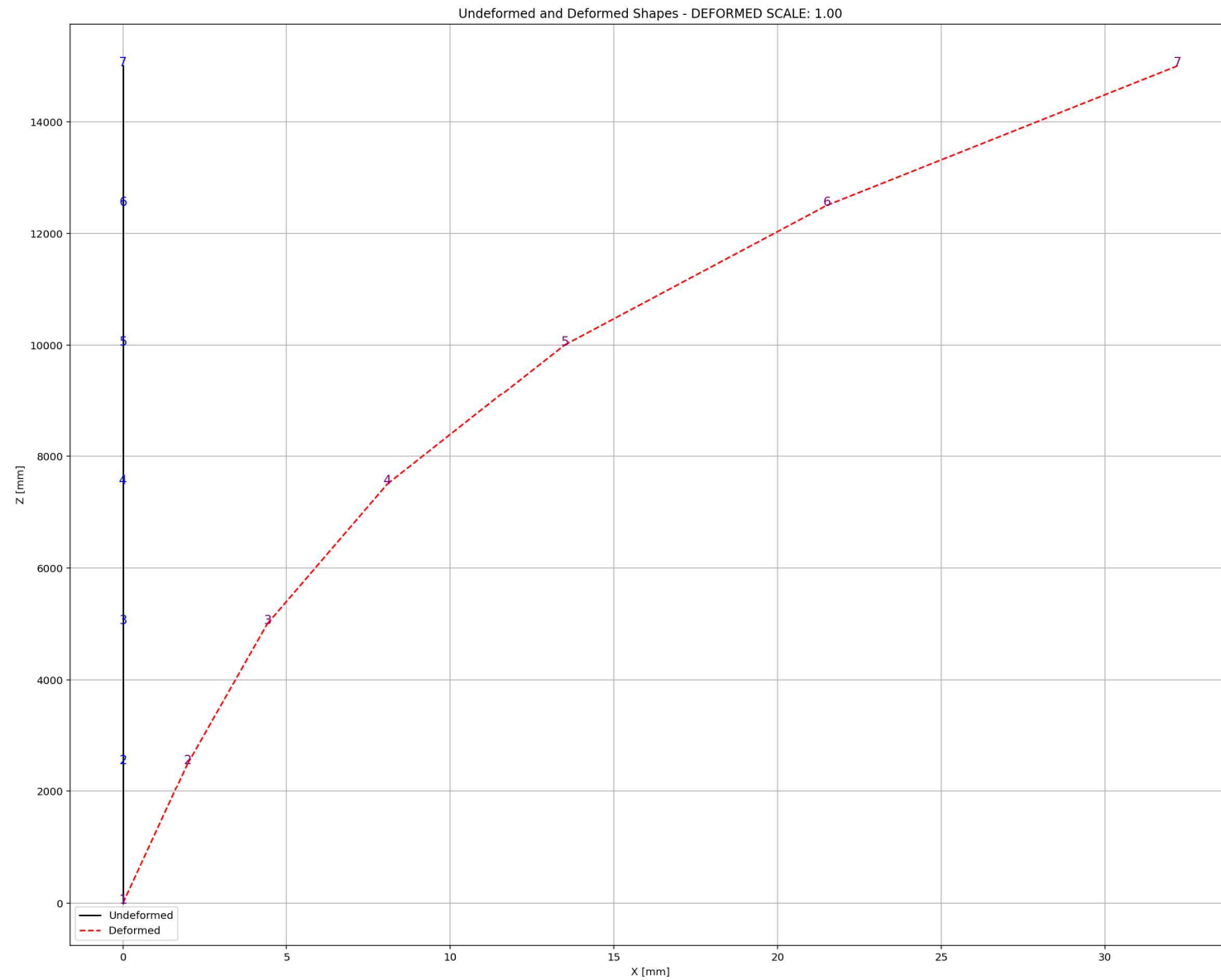
Period of Structure During Seismic Analysis



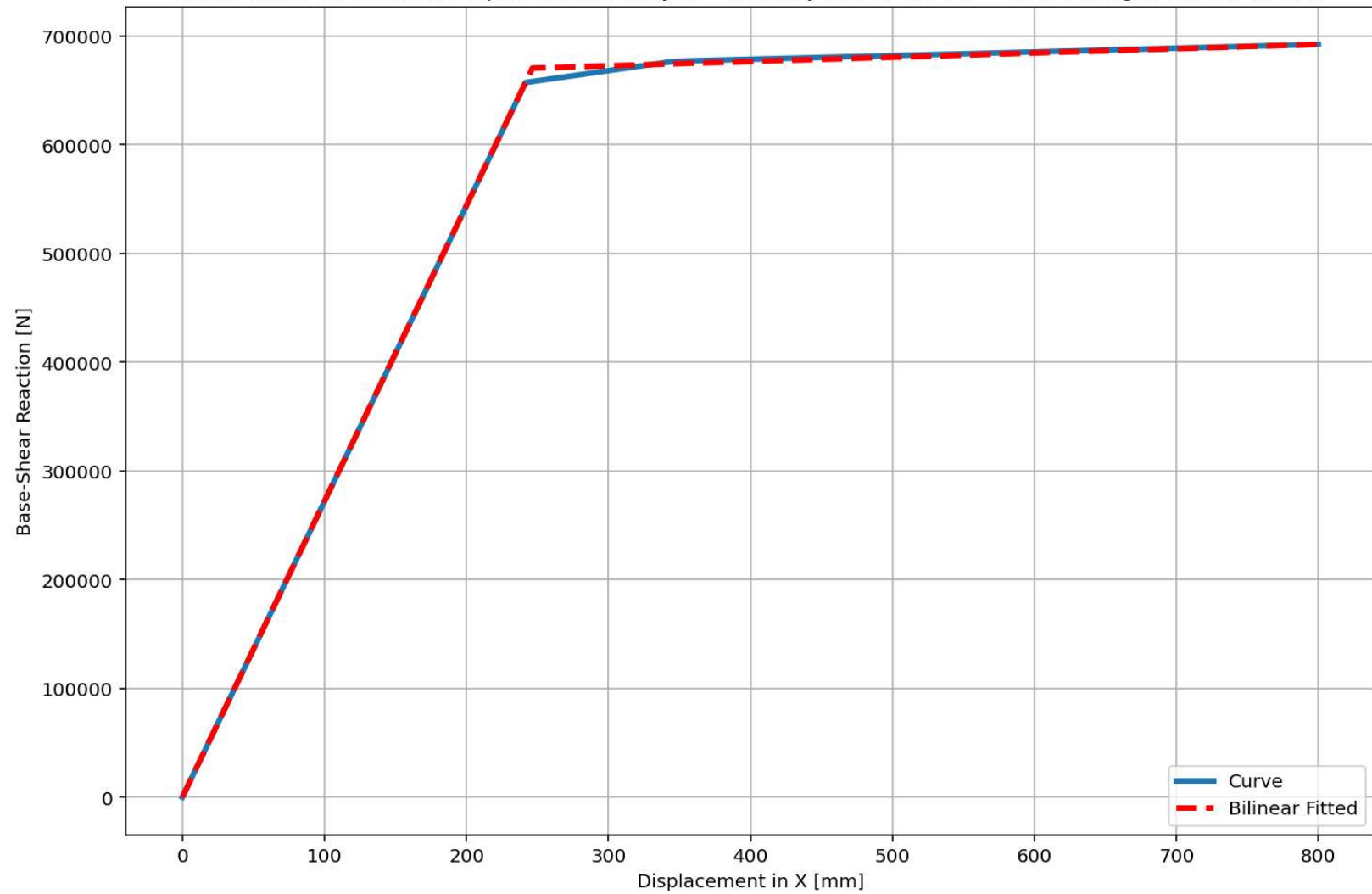
Behavior of beam - Stress-strain of Top and Bottom Fibers Curve







Last Data of Base Shear-Displacement Analysis - Ductility Ratio: 3.2459 - Over Strength Factor: 1.0325



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[2.46467312e+02 6.70580800e+05]

[8.00000000e+02 6.92349023e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 2720.77

Structure Plastic Stiffness : 865.44

Structure Tangent Stiffness : 39.33

Structure Ductility Ratio : 3.25

Structure Over Strength Factor: 1.03

+-----+

Over Strength Coefficient (Ω_0): 1.0325

Displacement Ductility Ratio (μ): 3.2459

Ductility Coefficient (R_μ): 3.2459

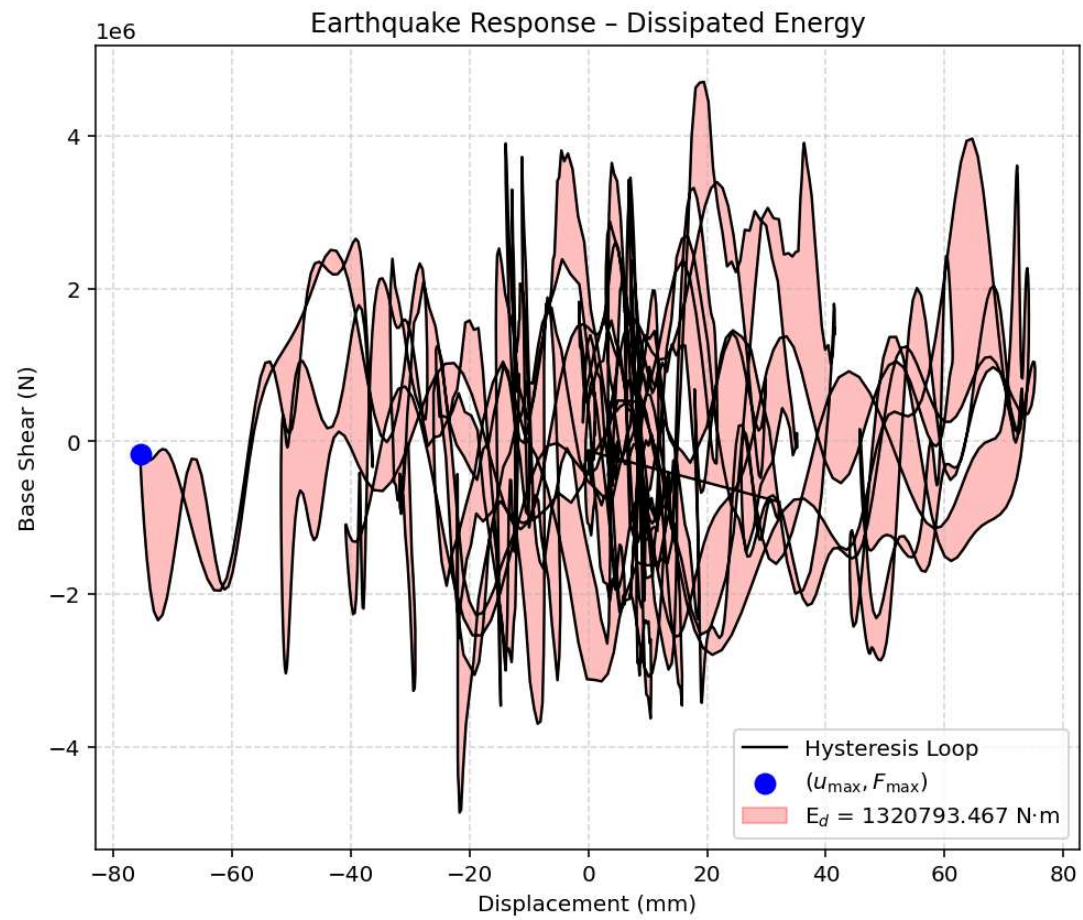
Structural Behavior Coefficient (R): 3.3512

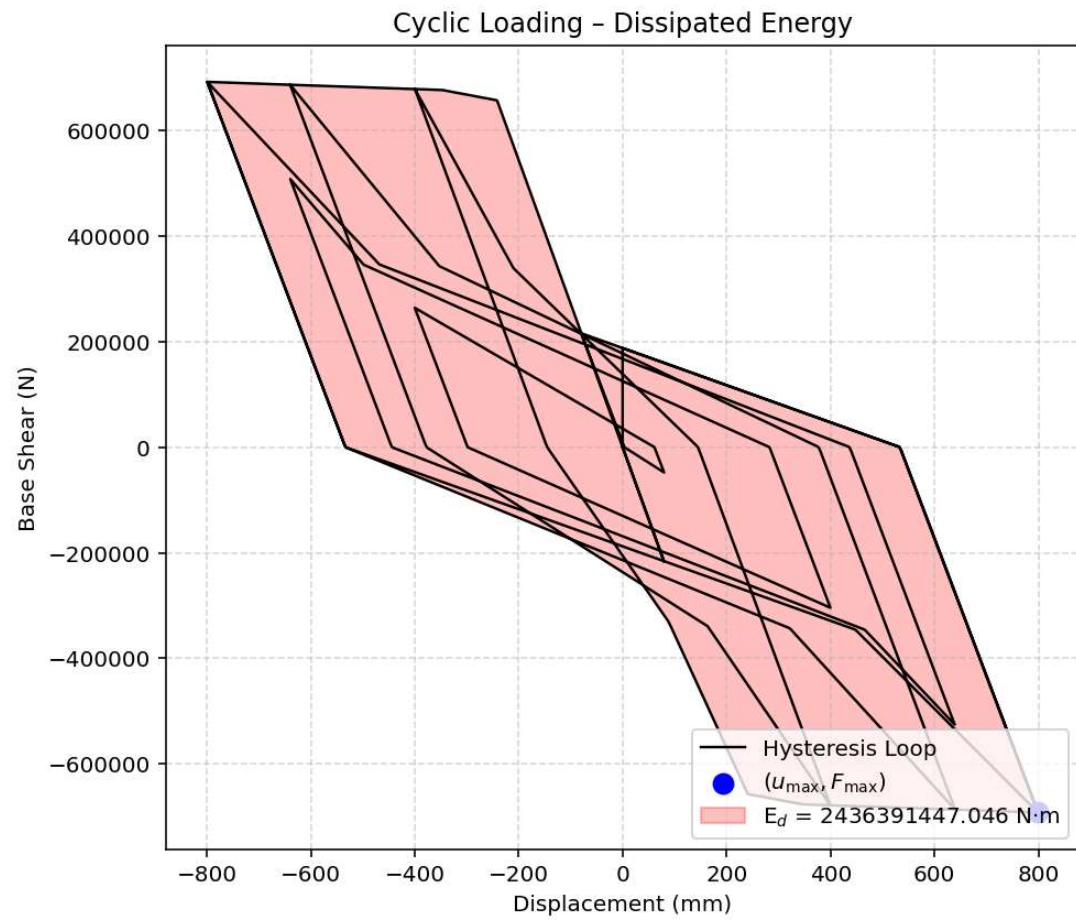
Structural Ductility Damage Index in X Direction: -30.8833 (%)

Dissipated Energy from Earthquake= 1012798067.08 N·mm

Dissipated Energy from Cyclic Displacement= 1085574700.52 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 9.330e+01 [%]





Dissipated Energy from Earthquake= 1320793.47 N·mm

Dissipated Energy from Cyclic Displacement= 2436391447.05 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.054 [%]

ZONE 1: VERY MINOR DAMAGE

